

Employment Preferences of *Favela* Residents

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Abstract

We study how workers at the margin of informality value job amenities and how these preferences shape sorting across sectors in a large Brazilian favela. We combine rich open-ended survey questions with a discrete choice experiment fielded to 700 young residents of Rio de Janeiro's Complexo da Maré. Open-text responses show that workers place formal benefits at the top of their priority ranking, followed by learning opportunities, good pay, and schedule flexibility, while complaints about past jobs center on management quality, difficult customers, and inflexible schedules. In the experiment, respondents are willing to forgo roughly one-quarter of wages for unemployment insurance, parental leave, and learning opportunities, but exhibit near-zero willingness to pay for termination notice or shorter commutes, despite demanding large wage premia to relinquish them. These gaps between willingness to pay and willingness to accept persist after conditioning on individual fixed effects, consistent with reference-dependent preferences rather than pure sorting. Valuations are heterogeneous by gender and employment status: formal workers value formal benefits most, the self-employed value them little, and the unemployed value insurance and learning opportunities most. Our results highlight the role of worker preferences over non-wage amenities in sustaining dual labor markets.

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1 Introduction

Labor markets in developing countries have large formal and informal sectors whose firms have substantial labor market power (Ulyssea, 2020; Felix, 2022; Amodio, Medina and Morlacco, 2025; Amodio et al., 2025). Why? Considerable research examines the demand side of informality: How firms choose whether to formalize and whether to hire workers informally (e.g., Ulyssea, 2018a). Far less is known about its supply side, even though worker preferences over non-wage attributes – which firms can exploit to pay workers less than their marginal revenue product – are the key link between the two phenomena. What do workers at the margin of informality value at work? Do they sort into informal wage work and self-employment based on these preferences? And to what extent does a mismatch between worker preferences and the attributes of formal jobs, many enshrined in labor codes, explain the dual nature of developing countries’ labor markets?

We answer these questions with novel survey data on work preferences of residents from Brazil’s largest *favela* complex. Recent work emphasizes that workers value job amenities beyond wages, including meaning and purpose (Ashraf et al., 2025) and scheduling flexibility (Mas and Pallais, 2017), but this evidence is predominantly from developed countries. While existing models of informality emphasize compensating differentials between formal and informal employment (Meghir, Narita and Robin, 2015; Haanwinckel and Soares, 2021), direct evidence on worker valuations at the margin remains scarce. We combine open-ended questions with discrete choice experiments to estimate willingness to pay for job amenities.

We begin by documenting four descriptive facts from open-text answers. First, workers mention formal employment benefits as their highest priorities, followed by learning and growth opportunities, and remuneration. Second, when asked about dream jobs, workers emphasize salary, benefits, and professional growth. Third, workers’ primary complaints about current or past jobs center on poor management, difficult customers, and inflexible schedules. Fourth, while workers cite flexibility and autonomy as attractive features of self-employment, lack of money emerges as the primary barrier preventing them from pursuing it.

We then present an experimental design to elicit preferences and estimate willingness to pay for specific job amenities of formal employment (i.e., unemployment insurance, parental leave, and termination notice) and for highly ranked attributes that are not specific to formal employment (i.e., learning opportunities and commute time). We designed the survey experiment to minimize known challenges when eliciting preferences (e.g., social desirability bias) and to mitigate the effects of behavioral biases (e.g., reference dependence) that might be particularly acute in developing countries. Each respondent completed five choice tasks presented in random order. In each task, respondents chose between two hypothetical job offers for the same position, with identical tasks and responsibilities. The offers differed only in one non-wage attribute and the wage.

Our findings from the discrete choice experiment show large and systematic valuations for some job amenities and essentially none for others. Favela residents are willing to give up roughly one-quarter of wages for unemployment insurance, parental leave, and learning opportunities, but they place near-zero willingness to pay on termination notice or shorter commutes, despite requiring large wage premia to forgo them—evidence consistent with reference-dependent preferences. These patterns are heterogeneous: women value parental leave far more than men, while all groups value unemployment insurance highly. We also find sorting across

sectors: formal workers place the highest monetary value on formal sector benefits, the self-employed value them little, and the unemployed value insurance and learning opportunities most. Together, the results suggest that both reference points and sectoral selection shape how workers price non-wage amenities.

Overall, our findings reveal that workers highly value formal job characteristics yet simultaneously desire flexibility, with financial constraints preventing access to arrangements that satisfy both preferences. This evidence complements existing work on firm behavior and enforcement policies by documenting how worker preferences shape labor market equilibrium at the margin of informality.

2 Context and Research Design

Labor markets in Brazil. Brazil has the sixth largest labor market in the world, with over 100 million people in its workforce (World Bank, 2025), and substantial informality characterizes it: 38% of workers are in the informal sector (IBGE, 2025). Formal employment in Brazil requires a signed work card (*carteira de trabalho*) that registers a worker’s employment history and guarantees access to labor rights (Ulyssea, 2018b). These rights include paid vacation, a thirteenth salary (annual bonus), severance protection, social security contributions toward retirement, and regulated working hours. Workers without a signed *carteira* operate as informal workers and lack these protections. However, informal employment typically offers greater flexibility of schedule and hours.

Favelas in Rio de Janeiro. Brazilian favelas are urban neighborhoods of dense informal settlements, home to around 1.5 million people in Rio de Janeiro alone. Table A.1 compares favela and non-favela regions in Rio using data from the 2010 Brazilian census. Favela residents are less likely to be literate (84% inside favelas vs. 92% outside) or self-identify as not Black (33% vs. 57%), and have nearly four times lower per capita income (R\$383 vs. R\$1,376). Our study took place in *Complexo da Maré*, Brazil’s largest favela complex and home to 130,000 people. Table A.1 shows that Maré’s characteristics align closely with other Rio favelas, suggesting our findings can generalize to similar contexts.¹

Sample. We surveyed 700 Maré residents aged 18-30 who are working or actively seeking employment. We recruited participants through door-to-door household surveys, visiting all households building-by-building during weekday evenings and weekend afternoons. We collected responses from one eligible individual per household across three collection waves between September and November 2025. Table A.2 presents summary statistics for our sample. Among respondents, 76% are female with an average age of 24. The majority (75%) identify as Black,² and 74% were born in Maré. Over half (56%) have children, and 19% receive the cash transfer program *Bolsa Família*. Regarding employment, 61% are currently employed, with formal sector workers comprising 42% of the sample and informal sector workers comprising 31%; 17% are self-employed. About half (52%) are actively searching for jobs. Average monthly wages among those with

¹Limitations in publicly available Census data prevent us from comparing employment characteristics across favelas, non-favelas, and Maré. For details on Brazilian favelas and employment barriers in these regions, see Angeli, Matavelli and Secco (2023) and Campos-Vázquez et al. (2025).

²In Brazil, “Black” combines *preto* (Black) and *pardo* (mixed race) categories.

employment experience reach R\$1,574, closely aligned with the R\$1,606 reservation wage among those who never worked.

Survey. Our survey comprises four main blocks. First, we collect detailed employment history including current employment status, formality status, occupation, wages, hours worked, commute time, and job benefits. For those not currently employed, we ask about their most recent job, and for those who never worked, their reservation wages. Second, we administer comprehensive open-ended questions about job aspirations and preferences. These questions capture respondents’ dream job characteristics, what they value most in jobs beyond wages, difficult aspects of work, reasons for job transitions, and barriers to desired employment. We also ask respondents to rank eight job amenities by importance (benefits, remuneration, work location, payment method, work environment, contract type, work schedule, and initiative/autonomy) and assess their knowledge of formal employment benefits. Third, we implement a discrete choice experiment to elicit willingness to pay for job amenities. Fourth, we collect demographic and socioeconomic information including age, gender, race, place of birth, education, whether they have children, and receipt of government transfers. Appendix C.1 presents all the survey questions. Participants entered a prize draw for one of five R\$200 prizes ($\approx$ USD40) as compensation for their time.

We recruited and trained surveyors from the local community, who administered our door-to-door survey. This approach minimizes inattention and ensures respondents understand the questions. Interviewers received training on survey implementation and strategies to mitigate social desirability bias or experimenter demand effects, following best practices in the literature (see [De Quidt, Vesterlund and Wilson, 2019](#); [Bursztyn et al., 2025](#)). Interviewers provided neutral instructions, informing respondents only that the study concerned job preferences of young people in Brazil without emphasizing specific hypotheses. During the discrete choice module, interviewers handed the tablet directly to respondents, allowing them to provide their answers privately. This procedure promotes anonymous decision-making and reduces concerns about surveyor observation or evaluation, attenuating social desirability bias. We did not inform surveyors about the study’s hypotheses or expected effects, further limiting the scope for experimenter demand effects.

Open-ended questions. Following a growing economics literature, we elicit what is top of mind for workers by asking open-ended questions about employment preferences ([Haaland et al., 2024](#)). This method avoids leading participants toward answers based on researcher-determined categories. We instructed surveyors not to provide specific examples and to write down participants’ answers as close as possible to what they actually said. We asked five open-ended questions:

1. **Dream job:** “What would your dream job have or offer you? Tell us some characteristics your dream job would have.”
2. **Value in job:** “Besides salary, what else do you look for in a job?”
3. **Dislike in job:** “What do you like least in your current [last] job?”
4. **Reasons for self-employment:** “You said you would like to be self-employed. For what reasons would you rather be self-employed?”

5. **Barriers to self-employment:** “You said you would like to be self-employed. For what reasons are you not self-employed today?”

We analyze these open-text responses in two complementary ways. First, we present word clouds that visualize the most frequently mentioned terms after excluding common words such as connectors and articles to provide an overview of the raw data. Second, we systematically code all responses into mutually non-exclusive categories through qualitative analysis. This coding process involved reviewing samples of responses, defining category boundaries, and iteratively correcting miscategorizations until achieving classification rates of 87-98%. We employed AI tools to assist with pattern matching and initial categorization, but our research team validated all category definitions and coding rules. Appendix C.3 provides detailed category definitions, coding methodology, and validation procedures.

Discrete choice experiments. To estimate willingness to pay for different job amenities, respondents completed five choice tasks. The five attributes reflected benefits of formal employment in Brazil (unemployment insurance, parental leave, and termination notice) and highly valued attributes that are not specific to formal employment (learning opportunities and commute time). We randomized question order to address potential anchoring effects. To ensure that job options were realistic to respondents, “Job A” mirrored the respondent’s current or last job as in Mas and Pallais (2017), across all five experiments. In each experiment, we showed “Job A” side-by-side with “Job B”. The two jobs were identical except on two dimensions: one attribute and the wage. We told respondents that the two jobs entailed the same tasks and responsibilities. To minimize reference dependence or loss aversion, which might occur if respondents interpret choosing the “Job B” as losing their current job, we displayed all job options in neutral framing. The surveyors did not point out to respondents that “Job A” mirrored their current or last job. We also inform them that their answers are anonymous. Finally, to ensure that even very large valuations would be within the support of our experimentally-assigned wage differences, we used a large wage difference span, uniformly distributed between -30% to +30%.³ See Appendix C.2 for further details on the discrete choice experiment.⁴

3 Motivating facts

We present four facts about *favela* residents’ attitudes and preferences about work based on the open-text response questions of our survey.

Fact 1: Workers mention formal employment benefits as their highest priority, followed by learning and growth opportunities, and remuneration, but aggregate rankings mask heterogeneity across employment types.— When asked what they value most in a job, benefits (e.g., health insurance and meal vouchers) emerged as the top priority (31.3%) (Figure B.3a). Learning and professional growth opportunities came second (21.4%), and good pay, third (17.7%). Work-life balance (15.7%), career advancement (11.3%),

³On November 19, upon noticing that a substantial share of respondents always chose the same option, we increased that range to -60% to +60% to make sure we would be able to capture higher valuations within the wage change support.

⁴We excluded 50 responses from the discrete choice experiment analysis due to a coding error in the discrete choice questions, totaling N=650. Nevertheless, Table A.2 shows the discrete choice sample is similar to the full sample.

and good work environment (9.1%) followed. Frequently mentioned terms include “learning,” “experience,” “benefits,” “health,” “insurance,” “salary,” “growth,” “knowledge,” and “development” (Figure 1a). Figure B.1 further displays how workers rank eight job amenities from most to least important in a forced-ranking exercise. Benefits ranked first overall (chosen as most important by 22% of respondents), followed by remuneration (15%) and work environment (12.7%). These patterns mask substantial heterogeneity across employment types. Formal and informal workers prioritize benefits, remuneration, workplace environment and work location. In contrast, the self-employed rank remuneration only seventh, prioritizing instead benefits, work schedule, and form of payment, suggesting that those who choose self-employment value flexibility and autonomy (Figure B.2).

Fact 2: When describing their dream job, workers emphasize adequate salary, working in a desired profession, schedule flexibility, and formal benefits.— Workers emphasize good salary as the most important attribute of their dream job (45.7%) (Figure B.3b). Nearly one-third (31.7%) mentioned wanting to work in a specific profession or field, indicating aspirations beyond simply having any job. Schedule flexibility ranked third (19.1%), with workers mentioning Monday to Friday schedules, no weekend work, and ability to control their hours. Benefits such as meal vouchers and health insurance appeared in 18.7% of responses, while 14.3% mentioned job stability. Frequently mentioned terms include “salary,” “schedule,” “voucher,” “food,” “stability,” “health,” “benefits,” and “flexible” (Figure 1b). Beyond wages, meal vouchers ranks as the single most-cited benefit.

Fact 3: Workers’ primary complaints about their current or last jobs center on dealing with difficult clients or colleagues, schedules, and workload, highlighting tension between job demands and desired flexibility.— Figure 1c presents responses to what workers like the least about their current or most recent job. Workers most frequently mention “schedule,” “clients,” “workload,” “hours,” and “people.” Figure B.3c shows that management and organizational problems emerged as the top complaint (16.3%), including poor management, lack of support, being assigned multiple job functions, and poor communication. Difficult customers, clients, or patients ranked second at 15.7%, followed by inflexible schedules at 12.5%. Excessive work hours appeared in 11.9% of responses, while physical and mental exhaustion was mentioned by 8.4%. Notably, direct complaints about compensation appeared in only 6.2% of responses. This pattern contrasts with what workers value in a job. While workers prioritize benefits when evaluating jobs generally (Facts 1 and 2), their interactions with managers and customers and lack of flexibility are what make them dislike their jobs.

Fact 4: Workers desiring self-employment cite schedule flexibility and autonomy as primary motivations, but lack of financial capital presents an overwhelming barrier to entry.— Among those who want to be self-employed, nearly half (48.9%) cited schedule flexibility and the ability to make their own hours, often mentioning the need to balance work with childcare (Figure B.5a). Over one-quarter (26.7%) emphasized being their own boss and not having a supervisor, while 24.4% mentioned better income potential. Asked why they would like to be self-employed, workers cited words associated with flexibility and autonomy (“flexibility,” “schedule,” “own boss,” “money,” “time,” “being told what to do”) as well as family considerations (Figure B.4a). However, when asked why they are not currently self-employed, financial constraints

emerged as the main barrier (65.6%) (Figure B.5b). The other reasons are considerably less common: lack of professional experience (15.6%), and family care responsibilities (12.5%). Figure B.4b shows “money” and “lack” appearing far more frequently than any other terms, with many carrying similar meaning (“financial,” “income,” “capital,” “budget”).

4 Demand for job amenities

4.1 Preferences

Preferences. Respondent i derives indirect utility V_j^i over job j considered by her, which features wage w_j , amenity bundle Γ_j , and idiosyncratic taste ξ_j^i . Respondents derive log indirect utility $\beta \ln w_j$ from job j ’s wage, and indirect utility δ_a from each amenity a included in job j ’s amenity bundle. Specifically:

$$V_j^i = \beta \ln w_j + \sum_{a \in \Gamma_j} \delta_a \times (k_{aj} = 1) + \xi_j^i \quad (1)$$

Probability of choosing job with amenity. Following the discrete choice literature, we assume that the idiosyncratic taste parameters ξ_j^i are drawn i.i.d. from a Type-I Extreme Value distribution, yielding a closed-form expression for the probability that respondent i would choose a particular job j^1 over another j^0 as a function wages, attributes, and preference parameters. Consider two jobs that differ only in one amenity a^* and wages. Let a^* denote the experiment to value amenity a , j^1 the job that has the amenity, and j^0 the job that does not, such that $k_{a^*j^1} = 1$ and $k_{a^*j^0} = 0$. Then the probability that respondents choose job j^1 over job j^0 is given by:

$$P(V_{j^1} > V_{j^0}) = \frac{\exp[\beta \ln(w_{j^1}/w_{j^0}) + \delta_{a^*}]}{1 + \exp[\beta \ln(w_{j^1}/w_{j^0}) + \delta_{a^*}]} \quad (2)$$

Logit regression. Equation 2 specifies our logit regression on a sample of respondent $\times$ experiment pairs with five experiments per respondent, one for each amenity. The outcome variable is a dummy for whether the respondent chose the job with the amenity in the amenity’s respective experiment. The dependent variables are $\ln(w_{j^1}/w_{j^0})$ – which we randomize, identifying slope β – and five amenity dummies, whose order is also random, and which identify five δ_a intercepts. We cluster standard errors by respondent.

4.2 The value of an amenity

Indifference condition. Consider two jobs that offer the same amenities except one, a^* , and offer different wages. Let j^1 denote the job that has amenity a^* , and j^0 the job that does not, such that $k_{a^*j^1} = 1$ and $k_{a^*j^0} = 0$. Suppose also that $w_{j^1} < w_{j^0}$. That is, the job with amenity a^* pays a lower wage than the job without it. In this setting, amenity a^* ’s relative monetary value can be obtained by solving for wage ratio that would make respondent i indifferent between the two jobs. Setting $V_{j^1}^i = V_{j^0}^i$ gives:

$$\ln(w_{j^1}/w_{j^0}) = -\delta_{a^*}/\beta + (\xi_{j^0}^i - \xi_{j^1}^i)/\beta \quad (3)$$

Since by assumption the idiosyncratic tastes $\xi_{j^0}^i$ and $\xi_{j^1}^i$ are i.i.d., $\mathbb{E}[\xi_{j^0}^i - \xi_{j^1}^i] = 0$ in the population. This gives the log wage ratio at which respondents are (in expectation) indifferent between jobs j^1 and j^0 :

$$\ln\left(\frac{w_{j^1}}{w_{j^0}}\right) = -\frac{\delta_a^*}{\beta} \quad (4)$$

Monetary value of an amenity. Following [Maestas et al. \(2023\)](#),⁵ we derive the monetary value of amenity a^* (henceforth MV_{a^*}) as the value, in Brazilian reais, that makes respondents indifferent between a job with amenity a that pays $w_{j^1} = w_{j^0} - MV_{a^*}$ and a job that does not have amenity a^* and pays $w_{j^0} = w_{j^1} + MV_{a^*}$:

$$\ln\left(\frac{w_{j^0} - MV_{a^*}}{w_{j^0}}\right) = -\frac{\delta_a^*}{\beta} \quad (5)$$

To facilitate comparison with reference-dependent valuations defined below, we adopt the convention of expressing MV_{a^*} as a (positive) fraction of w_{j^0} . We define and report RMV_{a^*} , the monetary value of amenity a^* relative to an identical job without it, as:⁶

$$RMV_{a^*} \equiv \frac{MV_{a^*}}{w_{j^0}} = 1 - \exp\left(-\frac{\delta_a^*}{\beta}\right). \quad (6)$$

4.3 Generalizing preferences to allow for reference dependence

The preference structure in equation 1 implies that respondents value amenities equally regardless of whether they already have them or not (see Appendix C.3). In this case, RMV_{a^*} equals respondents' willingness to pay (henceforth WTP_{a^*}) in forgone wages to get amenity a^* , and it also equals respondents' willingness to accept (henceforth WTA_{a^*}) in extra wages to forgo amenity a^* . However, WTP_{a^*} and WTA_{a^*} need not coincide if preferences for amenity a^* are reference-dependent.

To allow for the possibility of reference dependence in preferences, we generalize equation 1 to allow for amenity-specific valuations that differ depending on whether the respondent has the amenity *in their current job*.⁷ Let $V_{ja^*}^i$ denote the indirect utility derived by respondent i for job j in experiment for amenity a^* . Ignoring the indirect utility derived from other amenities, since they are identical across the two job options in experiment a^* , we have:

$$V_{ja^*}^i = \beta \ln w_j + \delta_{a^*} \times (k_{a^*j} = 1) + \underbrace{(\gamma \ln w_j + \theta_{a^*})(k_{a^*j} = 1 = k_{a^*r^i})}_{\text{reference dependence}} + \xi_{a^*j}^i \quad (7)$$

where r^i is the reference (i.e., current) job.

⁵ MV_{a^*} in our paper corresponds to “wage-increase equivalent” in [Maestas et al. \(2023\)](#)’s terminology.

⁶ To see this, solve for MV_{a^*} to get $MV_{a^*} = w_{j^0}(1 - \exp[-\delta_{a^*}/\beta])$.

⁷ While participants are not informed of this design choice, “Job A” (or the baseline job) mimics their current job, as in [Mas and Pallais \(2017\)](#). We therefore directly observe if those who tend to choose the job with the amenity are disproportionately those for which “Job A” has the amenity.

Extended logit regression. We estimate the preference parameters in equation 7 by modifying the original logit regression specified by equation 2 to interact the random wage gap and the amenity dummies with indicators for whether the respondent has the amenity in their current job.

4.3.1 Reference dependence versus sorting

One important reason why *estimates* of WTP_{a^*} and WTA_{a^*} need not coincide is sorting/selection into current jobs. If respondents sort into jobs that have the amenities they value most, we would find that those that currently have the amenity value it more, while those that do not have it value it less. We attempt to disentangle reference dependence from sorting by augmenting the preference structure posited in equation 7 to include a respondent taste shifter ξ^i :

$$V_{ja^*}^i = \beta \ln w_j + \delta_{a^*} \times (k_{a^*j} = 1) + \underbrace{(\gamma \ln w_j + \theta_{a^*})(k_{a^*j} = 1 = k_{a^*ri})}_{\text{reference dependence}} + \underbrace{\xi^i}_{\text{Respondent taste shifter}} + \xi_{a^*j}^i \quad (8)$$

Respondent's taste shifter ξ^i —in practice added in the logit estimation as a respondent fixed effect—helps to disentangle amenity-specific reference dependence from sorting by restricting the variation used to estimate all other preference coefficients to within-respondent, cross-experiment variation. That is, it captures respondents' *mean valuation over the amenity bundles* she encounters across all five experiments. If respondents sort into their current jobs based on their expected value of those jobs' amenity bundles, then preference estimates conditional on respondent fixed effects deliver a closer test of amenity-specific reference dependence.

Conditional logit regression. We estimate the preference parameters in equation 8 by adding a respondent fixed effect to the extended logit regression described earlier. Importantly, note that conditioning estimates on a respondent fixed effect changes the interpretation of all valuation metrics to be *relative to the omitted amenity* a^o . To make this explicit, we use notation RMV_{a^*/a^o} , WTP_{a^*/a^o} , and WTA_{a^*/a^o} for estimates conditional on respondent fixed effects. For example, WTP_{a^*/a^o} is respondents' willingness to pay for amenity a^* —in a job that offers neither a^* nor a^o —relative to their willingness to pay amenity a^o . Also, because the preference parameters that underlie WTP and WTA are jointly estimated, they share the same omitted category, which we set as WTP_{a^o} for ease of interpretation.

We use *shorter commute* as the omitted amenity for the main results—an amenity for which find positive but lower valuations than other amenities for all demographic sub-samples, thus facilitating the interpretation of relative coefficients. We report estimates omitting other amenities in the Appendix.⁸

Note that the identification of WTP_{a^*/a^o} is driven by respondents whose current jobs have neither amenity

⁸ Another attractive feature of omitting shorter commute is that its underlying attribute—commute time, a variable with 5 different categories indicating ranges (e.g., 30min, 1-hour, etc.)—was randomized in the alternative job (“Job B”) regardless of its value in respondents' baseline job (“Job A”) and the wage change was also random in either positive or negative direction. This further helps disentangle presence of the amenity in the baseline job (the shorter commute job might be the alternative job rather than the baseline job) from preference for a shorter commute, as the shorter commute option is the alternative for many respondents. Unlike other amenities, where its presence is a clear “benefit”, it was not ex ante clear to us whether a shorter commute was preferable in a slum context, since some people might prefer being away from the slum for most of the day. Despite our entertaining of this possibility, our findings suggest that slum residents value shorter commutes, though by less than they value other amenities.

a^* nor amenity a^o , while WTA_{a^*/a^o} is driven by respondents whose current jobs have a^* but not a^o .

4.4 Heterogeneity by demographics

To understand whether amenities are valued differently across demographic groups, we report separate relative monetary values for women versus men and for Black (“preto ou pardo”) versus non-Black slum residents. These estimates are based on separate logit regressions for each demographic sub-sample.

5 Experimental results

Attention sample (“switchers”). To ensure that our estimates are not biased by inattention (see Section 5.1 for a detailed discussion), our analysis focuses on the 434 respondents (67% of the sample) who sometimes chose the option with the amenity, sometimes the option without it. We refer to these as “switchers”. Our rationale is that switchers pay close attention to the wage differentials. Appendix Table A.3 shows that switchers’ demographic composition and distribution across employment sectors is nearly identical to the overall sample: 41% are in formal wage work, 32% in informal wage work, 18% self-employed, and 9% unemployed. On demographics, 74% are women, 72% are Black.⁹ We report estimates for the full sample in Appendix Table A.8. All estimates are larger in the full sample than in the attention sample, as expected.

Visual evidence. Figure 2 displays the underlieing experimental variation that identifies the demand parameters in equation 1, on which all *RMV* estimates are based. Demand for amenities is downward sloping for all amenities. Figure 3 displays the experimental variation that identifies the preference parameters in equation 7, on which estimates of *WTP* and *WTA* without fixed effects are based. Respondents whose current job has the amenity consistently pick the job with the amenity (in that amenity’s respective experiment). Similarly, respondents whose current job does not have the amenity more frequently choose the job that does not have it. Figure 4 displays the experimental variation that identifies the preference parameters in equation 8, on which all estimates conditional on respondent fixed effects are based. Amenity intercepts significantly decrease. The paragraphs below break down these effects into main findings.

Valuation estimates. Table 1 presents our estimates of relative monetary values, willingness to pay, and willingness to accept, for the five amenities in our discrete choice experiment (i.e., unemployment insurance, parental leave, termination notice, and learning opportunity). Columns (1) present results from preference parameters based on equation 1, columns (2)-(3) present estimates based on the preference parameters that allow for reference dependence (i.e., from equation 7, and columns (4)-(6) do the same but conditional on respondent fixed effects. Table 2 shows *WTP* estimates for the five amenities separately by demographic groups. Table 3 shows *RMV* estimates separately by respondents’ current employment status.

Finding 1: High valuations for some amenities, none for others. On average, slum residents are willing to pay up to 24.8% in wages to obtain unemployment insurance, 20.7% for parental leave, and 23.2% for learning opportunity. On average, we find near-zero and statistically insignificant willingness to pay for either

⁹The corresponding stats for the full sample are: 42% are in formal wage work, 31% in informal wage work, 17% self-employed, 10% unemployed, 77% women, 74% Black.

termination notice or shorter commute. Termination notice is the only of the three formal sector amenities we evaluate for which we find zero willingness to pay on average (columns (2) and (5) of Table 1) and across all demographic groups (Table 2). We also find near-zero willingness to pay for shorter commute across all demographic groups. However, willingness to accept to forgo both amenities is high: 30% for termination notice and 22% for shorter commute. Conditioning the estimation on respondent fixed effects—yielding estimates that are relative to the near-zero valuation on shorter commute—gives very similar results.

Finding 2: reference dependence. Columns (2) and (3) of Table 1 show that those amenities more than respondents who do not have them. The gap between the two valuations is the highest for unemployment insurance: favela residents require at least a 44.7% wage increase to forgo unemployment insurance but are only willing to pay 24.8% in wages if they do not have it. The other amenity for which gaps are substantial is shorter commute: residents require at least a 22% wage increase to accept a job with longer commute but are not willing to pay for one with a shorter commute. These gaps remain even after conditioning on individual fixed effects, consistent with reference-dependent preferences as opposed to sorting.

Finding 3: Heterogeneous valuations by demographics. Table 2 shows that all groups are willing to pay a substantial share of wages to obtain unemployment insurance, and no groups are willing to pay anything to obtain termination notice. Men are willing to pay up to 18.5% for unemployment insurance, and women up to 27%. Valuations are more heterogeneous for the other amenities. Women are willing to pay up to 28.2% in wages to have parental leave, whereas men are not willing to pay for that amenity at all. Appendix Table A.6 reports the same heterogeneity by willingness to accept. Consistent with reference dependence, *WTA* is higher than *WTP* for all amenities and groups.

Finding 4: Sorting on preferences across sectors. Table 3 presents evidence consistent with sorting across employment sectors (i.e., formal wage work, informal wage work, self-employment, and unemployment) on the amenities those sectors typically offer. Formal sector workers value amenities typically provided by the formal sector—i.e., unemployment insurance (40.8%), parental leave (24.5%), and termination notice (18.3%)—the most, whereas the self-employed see near-zero and statistically insignificant monetary value in either unemployment insurance or termination notice. The self-employed value parental leave highly (16.4%), but do so by less than all other employment sectors (valuations ranging between 24.3% and 28.3%), including the unemployed.¹⁰ Workers in informal wage work value termination notice the least (0.09%, significant at 10% level), but value shorter commute (17.3%) more than all other categories. This is consistent with job availability in the favela being primarily informal. Interestingly, the unemployed value unemployment insurance (46.8%) and learning opportunity (50%) the most across all groups.

5.1 inattention

Despite us conducting surveys in-person to maximize attention, the survey module with the discrete choice experiment is answered in private by respondents (the surveyors hand them the tablet). This was a conscious

¹⁰Importantly, while the self-employment sector in Brazil tends to be more composed by women than other sectors, the differences in *RMV* we document are unlikely to be driven by demographic preference heterogeneity because there is little variation in demographic composition across sectors in our survey sample (see Appendix Tables A.4 and A.5).

decision to minimize experimenter biases (i.e., the respondent thinking that the surveyor wants them to choose a specific option). Anticipating that respondents might not pay attention on how the two jobs differ in each experiment, we displayed these options in grid format to highlight the differences and the similarities between the two options. Nevertheless, inattention can affect decisions. Inattentive respondents might not notice the wage differentials, especially if they do not differ in the left-most digit, for example, but more easily notice a “Yes” for the tested amenity in the job that has it. These participants might therefore always choose the option with the amenity.

We find evidence consistent with high levels of inattention: 31% of participants always choose the job the amenity, even though they are shown similar wage gaps (random). We refer to these respondents as “always-takers”. Only 12 participants (1.8%) never chose the job with the amenity. We refer to these as “never-takers”. Of course, it is possible that “always takers” are not inattentive, but instead have much larger valuations than spanned by our 0-30% randomly assigned wage gap.¹¹ Their behavior might also be partially driven by reference dependence, a phenomenon we analyze separately in our “attention sample”. In fact, Appendix Table A.3 shows that most always-takers are formally employed (45%), while most never-takers are self-employed (50%). Since three of the amenities we evaluate are primarily present in the formal sector (unemployment insurance, termination notice, and parental leave), this suggests that the behavior of always-takers can be driven by either inattention or reference dependence.

6 Discussion

Our findings speak to the supply-side foundations of dual labor markets in developing countries. Favela residents place very high monetary value on unemployment insurance, parental leave, and learning opportunities, indicating that core features of formal jobs are strongly desired at the margin of informality. At the same time, they are unwilling to sacrifice wages for termination notice or shorter commutes, even though they demand large wage premia to give these up, a pattern consistent with reference-dependent preferences rather than simple compensating differentials.

Heterogeneity by gender and race, and especially across sectors, shows that workers sort into formal jobs, informal wage work, and self-employment partly because of these valuations: formal workers value formal sector benefits most, the self-employed appear to trade insurance for autonomy and schedule control, and the unemployed attach the highest value to insurance and skill acquisition. Taken together, the results suggest that worker preferences over non-wage attributes, shaped by reference points and sector-specific constraints, help sustain the coexistence of formal and informal employment.

These patterns have direct implications for policy design and for models of informality. First, they suggest that formal labor codes in Brazil bundle together attributes that workers value very differently: reinforcing unemployment insurance and parental leave may generate large welfare gains, whereas rigid termination rules appear less salient ex ante and operate mainly through loss aversion once acquired.

¹¹Towards the end of the survey, on November 19, upon seeing a significant share of always-takers, we increased this range to 0 to 60%.

Second, the tension between strong demand for formal benefits and a revealed preference for flexibility and autonomy—combined with binding financial constraints to self-employment—highlights that expanding access to “formal-style” social protection outside standard contracts and easing capital constraints for small entrepreneurs may be more effective than simply tightening enforcement of existing regulations. More broadly, our evidence calls for models in which informality is not only a response to enforcement and firm costs, but also an outcome of heterogeneous, reference-dependent worker preferences over bundled amenities, flexibility, and risk-sharing.

References

- Amodio, Francesco, Emanuele Brancati, Peter Brummund, Nicolás de Roux, Michele Di Maio, et al.** 2025. “Labor Market Power and Self-Employment Around the World.” Rockwool Foundation Berlin (RF Berlin).
- Amodio, Francesco, Pamela Medina, and Monica Morlacco.** 2025. “Labor Market Power, Self-Employment, and Development.” *American Economic Review*, 115(9): 3014–3057.
- Angeli, Deivis, Ieda Matavelli, and Fernando Secco.** 2023. “Expected Discrimination and Job Search.” Unpublished Manuscript.
- Ashraf, Nava, Oriana Bandiera, Virginia Minni, and Luigi Zingales.** 2025. “Meaning at Work.” NBER Working Paper No. 33843, R&R at Quarterly Journal of Economics.
- Bursztyn, Leonardo, Ingar K. Haaland, Nicolas Röver, and Christopher Roth.** 2025. “The Social Desirability Atlas.” National Bureau of Economic Research Working Paper 33920.
- Campos-Vázquez, Raymundo M, Robert Duval-Hernández, Laura Juárez, and M Ehekatzin García-Guzmán.** 2025. “Do Workers Value Formal Jobs? A Discrete Choice Experiment in Mexico.”
- De Quidt, Jonathan, Lise Vesterlund, and Alistair J Wilson.** 2019. “Experimenter Demand Effects.” *Handbook of research methods and applications in experimental economics*, 384.
- Felix, Mayara.** 2022. “Trade, Labor Market Concentration, and Wages.” Unpublished Manuscript.
- Haaland, Ingar, Christopher Roth, Stefanie Stantcheva, and Johannes Wohlfart.** 2024. “Measuring what is top of mind.” ECONtribute Discussion Paper.
- Haanwinckel, Daniel, and Rodrigo R Soares.** 2021. “Workforce Composition, Productivity, and Labour Regulations in a Compensating Differentials Theory of Informality.” *Review of Economic Studies*, 88(6): 2970–3010.
- IBGE.** 2025. “Pesquisa Nacional por Amostra de Domicílios Contínua – PNAD Contínua.” *Instituto Brasileiro de Geografia e Estatística*, Third Quarter 2025.
- Maestas, Nicole, Kathleen J Mullen, David Powell, Till Von Wachter, and Jeffrey B Wenger.** 2023. “The value of working conditions in the United States and implications for the structure of wages.” *American Economic Review*, 113(7): 2007–2047.
- Mas, Alexandre, and Amanda Pallais.** 2017. “Valuing Alternative Work Arrangements.” *American Economic Review*, 107(12): 3722–3759.
- Meghir, Costas, Renata Narita, and Jean-Marc Robin.** 2015. “Wages and Informality in Developing Countries.” *American Economic Review*, 105(4): 1509–1546.
- Ulyssea, Gabriel.** 2018a. “Firms, Informality, and Development: Theory and Evidence from Brazil.” *American Economic Review*, 108(8): 2015–2047.

- Ulyssea, Gabriel.** 2018*b*. “Firms, informality, and development: Theory and evidence from Brazil.” *American Economic Review*, 108(8): 2015–2047.
- Ulyssea, Gabriel.** 2020. “Informality: Causes and Consequences for Development.” *Annual Review of Economics*, 12(Volume 12, 2020): 525–546. Publisher: Annual Reviews.
- World Bank.** 2025. “Labor Force, Total.” World Development Indicators [Data file].

Main Tables and Figures

Table 1: Relative monetary values, willingness to pay for, and willingness to accept to forgo job amenities

	Within-respondent					
	Has in current job? (0/1)			Has in current job? (0/1)		
	(1) RMV_{a^*}	(2) WTP_{a^*}	(3) WTA_{a^*}	(4) RMV_{a^*}/a^0	(5) WTP_{a^*}/a^0	(6) WTA_{a^*}/a^0
Unemployment insurance	0.323*** (0.0230)	0.248*** (0.0350)	0.447*** (0.0382)	0.188*** (0.0308)	0.281*** (0.0480)	0.420*** (0.0400)
Parental leave	0.243*** (0.0229)	0.207*** (0.0392)	0.279*** (0.0293)	0.113*** (0.0336)	0.253*** (0.0478)	0.292*** (0.0390)
Termination notice	0.168*** (0.0220)	0.00515 (0.0494)	0.309*** (0.0342)	0.0365 (0.0360)	0.0780 (0.0566)	0.318*** (0.0422)
Learning opportunity	0.266*** (0.0221)	0.232*** (0.0389)	0.297*** (0.0321)	0.136*** (0.0321)	0.295*** (0.0461)	0.296*** (0.0387)
Shorter commute	0.146*** (0.0250)	-0.0479 (0.0675)	0.223*** (0.0315)	-	-	0.242*** (0.0434)
Respondent FE	No	No	Yes	Yes		
Observations	1980	1980	1980	1980	1980	

Note: This table shows estimates of relative monetary values (RMV), willingness to pay (WTP) for, and willingness to accept (WTA) to forgo job amenities. All estimates are expressed as percent of the wage for a job without the respective amenity. An observation is a respondent x amenity experiment pair. The sample is restricted to employed workers and excludes always-takers (i.e., respondents who chose always the job with amenity across all five amenity experiments) and never-takers (i.e., respondents who never chose the job with amenity across all five amenity experiments). RMV for each amenity in column (1) is the monetary value of the amenity, relative to the wage of job without the amenity. WTA for each amenity in column (2) is the percent wage increase respondents who have the amenity in their current job would be willing to accept to lose the amenity. WTP for each amenity in column (3) is the percent wage cut respondents who do not have the amenity in their current job would be willing to pay to receive the amenity. Estimates in column (1) are based on preference parameters estimated in a logit regression where the outcome variable is a dummy for whether the respondent chose the job with the amenity, and the explanatory variables are the random log wage gap (log wage of job with amenity / log wage of job without it), and five amenity dummies. Estimates for columns (2) and (3) are based on preference parameters from a single logit regression, where the random log wage gap and all amenity dummies are interacted with a dummy indicating whether the respondent has the respective amenity in their current job. Estimates for columns (4)-(6) add respondent fixed effects to the respective specifications in (1)-(3). Standard errors are obtained by cluster bootstrap at the respondent level (1,000 replications). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 2: Demographic heterogeneity in willingness to pay for job amenities

	(1) All	(2) Women	(3) Men	(4) Black	(5) Not Black
Unemployment insurance	0.248*** (0.0350)	0.270*** (0.0411)	0.185*** (0.0552)	0.243*** (0.0357)	0.276 (1.365)
Parental leave	0.207*** (0.0392)	0.282*** (0.0379)	-0.0312 (0.0880)	0.233*** (0.0343)	0.0347 (0.847)
Termination notice	0.00515 (0.0494)	0.0275 (0.0577)	-0.0616 (0.113)	0.0545 (0.0500)	-0.300 (10.75)
Learning opportunity	0.232*** (0.0389)	0.297*** (0.0409)	-0.00857 (0.107)	0.193*** (0.0370)	0.407* (0.241)
Shorter commute	-0.0479 (0.0675)	-0.0481 (0.0545)	-0.0501 (0.1000)	-0.0302 (0.0613)	-0.166 (1.211)
Observations	1980	1450	530	1450	530

Note: This table shows estimates of willingness to pay (WTP) separately by respondent demographics. See footnotes to Table 1. All estimates are expressed as percent of the wage for a job without the respective amenity. An observation is a respondent x amenity experiment pair. The sample is restricted to employed workers and excludes always-takers (i.e., respondents who chose always the job with amenity across all five amenity experiments) and never-takers (i.e., respondents who never chose the job with amenity across all five amenity experiments). The number of observations across demographic samples is the same by coincidence. The sample includes 1450 experiments for 290 women (210 Black, 80 not Black) and 530 experiments for 106 men (80 Black, 26 not Black). Standard errors are obtained by cluster bootstrap at the respondent level (1,000 replications). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

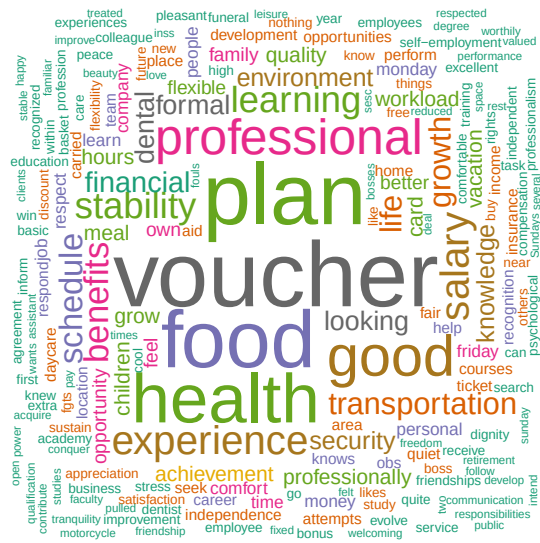
Table 3: Relative monetary value of job amenities by respondent employment status

	(1) All	(2) Formal	(3) Informal	(4) Self-employed	(5) Unemployed
Unemployment insurance	0.336*** (0.0222)	0.408*** (0.0435)	0.347*** (0.0439)	0.0887 (0.0599)	0.468*** (0.109)
Parental leave	0.243*** (0.0226)	0.245*** (0.0298)	0.283*** (0.0414)	0.164*** (0.0578)	0.243*** (0.0677)
Termination notice	0.183*** (0.0179)	0.258*** (0.0302)	0.0912* (0.0470)	0.0227 (0.0719)	0.322*** (0.0920)
Learning opportunity	0.285*** (0.0192)	0.270*** (0.0299)	0.259*** (0.0476)	0.270*** (0.0534)	0.509*** (0.112)
Shorter commute	0.118*** (0.0245)	0.121*** (0.0335)	0.173*** (0.0510)	0.169*** (0.0563)	-0.209 (0.389)
Observations	2170	885	705	390	190

Note: This table shows relative monetary values (RMV) for each amenity, as percent of the job without the amenity. Column (1) shows estimates including all respondents in the attention sample. Columns (2)-(5) restrict the sample to workers in formal employment, informal employment, self-employment, and unemployment, respectively. RMV for each amenity is the monetary value of the amenity, relative to the wage for the job without the amenity. An observation is a respondent x amenity experiment pair. RMV estimates are based on preference parameters estimated in a logit regression where the outcome variable is a dummy for whether the respondent chose the job with the amenity, and the explanatory variables are the log wage ratio between the job with the amenity and the job without it, and five amenity dummies. The attention sample excludes always-takers (i.e., respondents who always chose the job with amenity across all five amenity experiments) and never-takers (i.e., respondents who never chose the job with amenity across all five amenity experiments). Standard errors are obtained by cluster bootstrap at the respondent level (1,000 replications). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Figure 1: Open-text responses

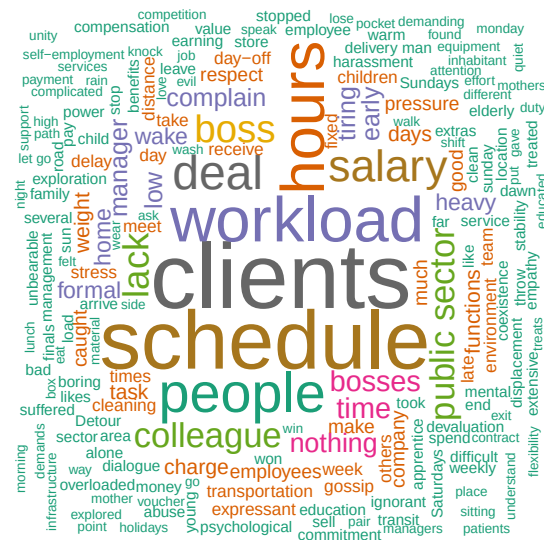
(a) Values in job



(b) Dream job

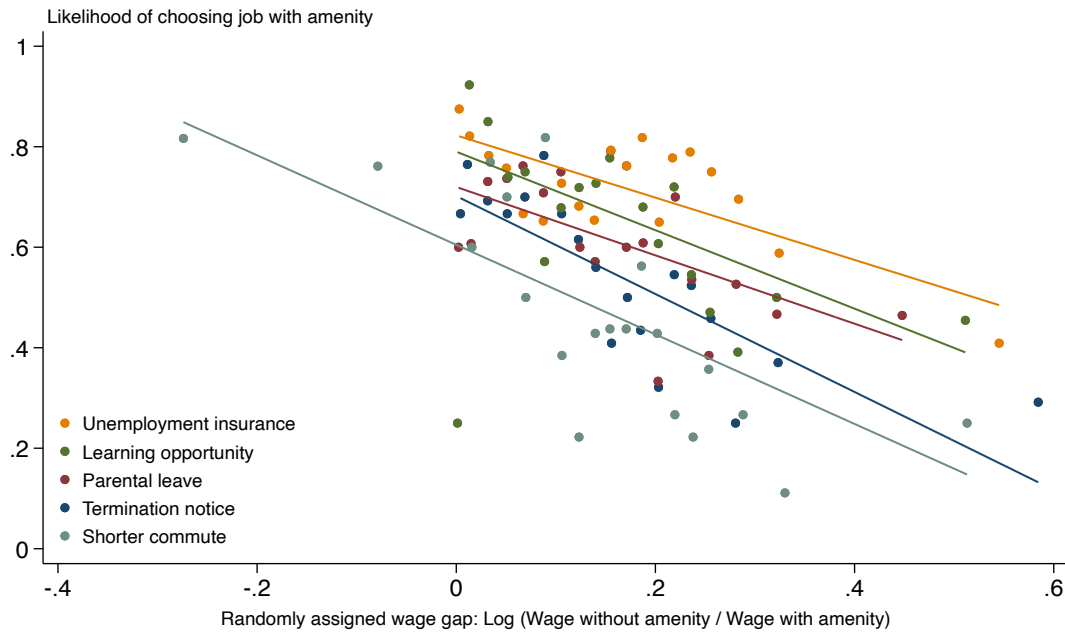


(c) Dislikes in job



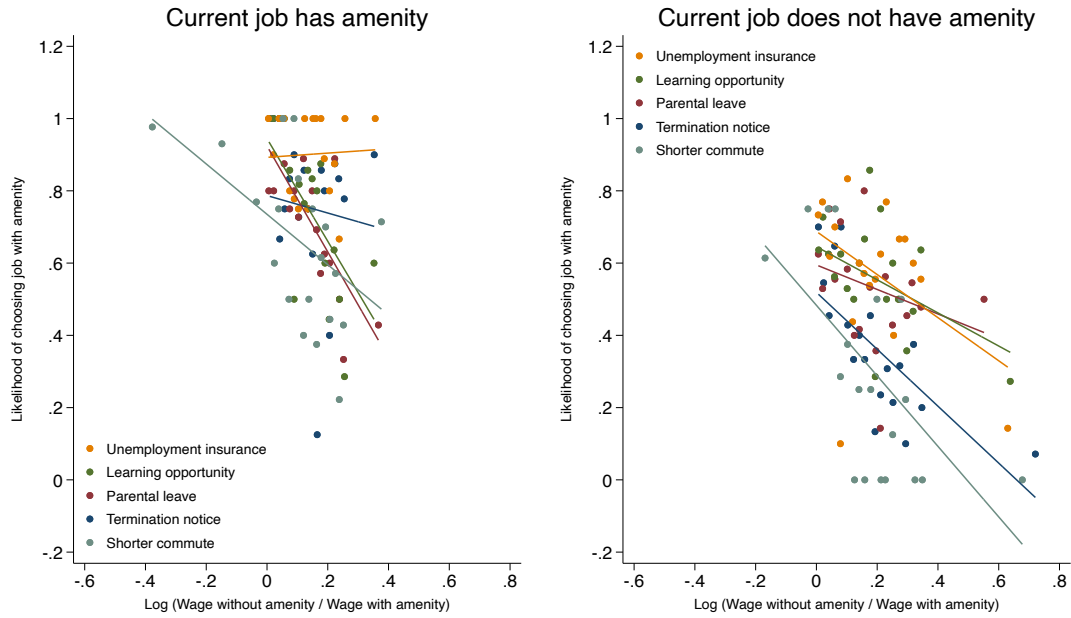
Notes: This figure plots word clouds with the responses to “What would your dream job have or offer you?” (N=700) (Panel a), “What did you like least about your most recent job?” (N=631) (Panel b), and “Besides salary, what else do you look for in a job?” (N=700) (Panel c). Word size reflects frequency.

Figure 2: Demand for job amenities



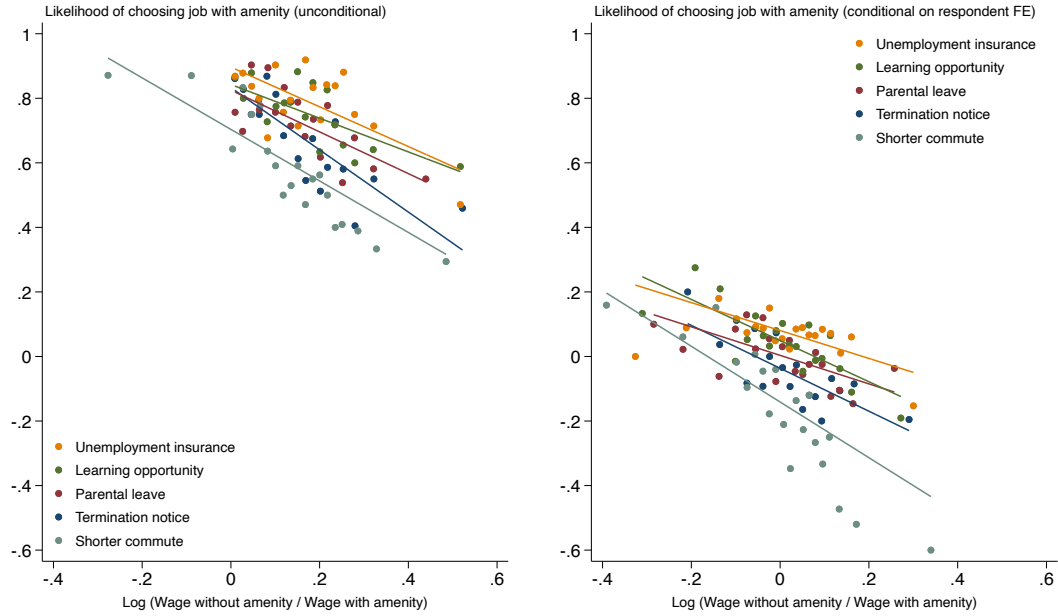
Note: This figure plots binned scatters, separately for each amenity experiment, of a dummy indicating if a respondent chooses the job option with the amenity in the experiment (y-axis) and the randomly assigned wage gap, reported as the log difference between the job with amenity and the job without. The sample excludes always-takers (i.e., respondents who chose always the job with amenity across all five amenity experiments) and never-takers (i.e., respondents who never chose the job with amenity across all five amenity experiments). See Appendix Table A.3 for descriptives on always-takers, never-takers, and switchers, and Appendix Figure B.6 for versions of the graph including those groups.

Figure 3: Reference-dependent demand by amenity status in current job



Note: This figure plots binned scatters, separately for each amenity experiment, and separately by whether the respondent reports currently having that amenity in their job, of a dummy indicating if a respondent chooses the job option with the amenity in the experiment (y-axis) and the randomly assigned wage gap, reported as the log difference between the job with amenity and the job without. The sample is restricted to employed workers and excludes always-takers (i.e., respondents who chose always the job with amenity across all five amenity experiments) and never-takers (i.e., respondents who never chose the job with amenity across all five amenity experiments). See Appendix Table A.3 for descriptives on always-takers, never-takers, and switchers, and Appendix Figure B.6 for versions of the graph including those groups.

Figure 4: Within-respondent demand for job amenities



Note: This figure plots binned scatters, separately for each amenity experiment, and separately by whether the axes are residualized on respondent fixed effects, of a dummy indicating if a respondent chooses the job option with the amenity in the experiment (y-axis) and the randomly assigned wage gap, reported as the log difference between the job with amenity and the job without. The sample is restricted to employed workers and excludes always-takers (i.e., respondents who chose always the job with amenity across all five amenity experiments) and never-takers (i.e., respondents who never chose the job with amenity across all five amenity experiments).

Online Appendix for
“Employment Preferences of *Favela* Residents”,
by Mayara Felix, Beatriz Marcoje, Ieda Matavelli, and Maria Clara Rodrigues.

Online Appendix A: Supplementary Tables

Table A.1: Rio de Janeiro summary statistics (2010 Census)

	Non-Favela	Favela	Maré
Population	4,888,663	1,391,953	129,715
Literacy Share	0.92	0.84	0.83
White Population Share	0.57	0.33	0.38
Income Per Capita (in 2010 Brazilian Reais)	1376.35	382.87	395.38

Note: This table shows summary statistics for neighborhoods in Rio de Janeiro municipality from the 2010 Brazilian Census. Favelas are neighborhoods defined as subnormal agglomerates (*aglomerados subnormais*) in the census classification. *Non-Favela* comprises all other census tracts. *Maré* comprises the favela complex where we conducted the data collection for this study.

Table A.2: Maré survey summary statistics (October-December 2025)

	Full Sample		Discrete Choice Sample	
	Mean	SD	Mean	SD
Female (0/1)	0.76	0.43	0.77	0.42
Age	23.82	3.74	23.89	3.74
High School (0/1)	0.65	0.48	0.64	0.48
Born in Maré (0/1)	0.74	0.44	0.74	0.44
Black (0/1)	0.75	0.43	0.74	0.44
Has Children (0/1)	0.56	0.50	0.56	0.50
Working (0/1)	0.61	0.49	0.62	0.49
Never Worked (0/1)	0.10	0.30	0.10	0.30
Searching for Job (0/1)	0.52	0.50	0.52	0.50
Formal Worker (0/1)	0.42	0.49	0.42	0.49
Informal Worker (0/1)	0.31	0.46	0.31	0.46
Self-Employed (0/1)	0.17	0.38	0.18	0.38
Reservation Wage (R\$)	1606.46	1123.63	1625.27	1173.26
Monthly Income (R\$)	1573.10	818.51	1595.85	827.70
Receives Cash Transfer (0/1)	0.19	0.40	0.20	0.40
Observations	700		650	

Note: This table shows summary statistics for our survey of N=700 respondents in Maré conducted between October-December 2025. The *Discrete Choice* sample excludes 50 observations due to a coding error in the discrete choice questions.

Table A.3: Job choice variation across respondents

	Always	Chooses job with amenity		Total
		Never	Sometimes	
N	203 (31.3%)	12 (1.8%)	434 (66.9%)	649 (100.0%)
Formally employed	0.45 (0.50)	0.17 (0.39)	0.41 (0.49)	0.42 (0.49)
Informally employed	0.29 (0.45)	0.33 (0.49)	0.32 (0.47)	0.31 (0.46)
Self-employed	0.14 (0.35)	0.50 (0.52)	0.18 (0.38)	0.17 (0.38)
Unemployed	0.12 (0.33)	0.00 (0.00)	0.09 (0.28)	0.10 (0.30)
Female	0.85 (0.36)	0.58 (0.51)	0.74 (0.44)	0.77 (0.42)
Black	0.80 (0.40)	0.67 (0.49)	0.72 (0.45)	0.74 (0.44)
Max random wage gap	0.19 (0.18)	0.45 (0.25)	0.25 (0.19)	0.23 (0.19)
Min random wage gap	-0.16 (0.12)	-0.13 (0.19)	-0.15 (0.13)	-0.15 (0.13)

Note: This table shows summary statistics for respondents who always, never, or sometimes choose the job with amenity across all five discrete choice experiments for respondents in the discrete choice experiment module. Black means “preto ou pardo”.

Table A.4: Employment status distribution by demographic groups

	Demographic group				Total
	Female, black	Female, not black	Male, black	Male, not black	
N	372 (57.3%)	126 (19.4%)	111 (17.1%)	40 (6.2%)	649 (100.0%)
sector					
Formal	156 (41.9%)	49 (38.9%)	46 (41.4%)	19 (47.5%)	270 (41.6%)
Informal	125 (33.6%)	38 (30.2%)	31 (27.9%)	9 (22.5%)	203 (31.3%)
Self-employed	60 (16.1%)	22 (17.5%)	23 (20.7%)	8 (20.0%)	113 (17.4%)
Unemployed	31 (8.3%)	17 (13.5%)	11 (9.9%)	4 (10.0%)	63 (9.7%)

Note: This table shows the distribution of current employment status by demographic groups for respondents in the discrete choice experiment module. Black means “preto ou pardo”.

Table A.5: Demographic distribution of employment sectors

	Employment sector				Total
	Formal	Informal	Self-employed	Unemployed	
N	270 (41.6%)	203 (31.3%)	113 (17.4%)	63 (9.7%)	649 (100.0%)
Female	0.76 (0.43)	0.80 (0.40)	0.73 (0.45)	0.76 (0.43)	0.77 (0.42)
Black	0.75 (0.43)	0.77 (0.42)	0.73 (0.44)	0.67 (0.48)	0.74 (0.44)

Note: This table shows the gender and race composition by current employment status for respondents in the discrete choice experiment module. Black means “preto ou pardo”.

Table A.6: Demographic heterogeneity in willingness to accept to forgo job amenities

	(1) All	(2) Women	(3) Men	(4) Black	(5) Not Black
Unemployment insurance	0.447*** (0.0382)	0.486*** (0.0592)	0.382*** (0.0552)	0.430*** (0.0508)	0.500*** (0.0837)
Parental leave	0.279*** (0.0293)	0.358*** (0.0532)	0.145*** (0.0533)	0.270*** (0.0319)	0.303*** (0.0724)
Termination notice	0.309*** (0.0342)	0.319*** (0.0491)	0.295*** (0.0590)	0.284*** (0.0430)	0.390*** (0.0738)
Learning opportunity	0.297*** (0.0321)	0.316*** (0.0490)	0.272*** (0.0561)	0.300*** (0.0373)	0.288*** (0.0922)
Shorter commute	0.223*** (0.0315)	0.242*** (0.0411)	0.190*** (0.0418)	0.206*** (0.0340)	0.282*** (0.0971)
Observations	1980	1450	530	1450	530

Note: This table shows estimates of willingness to accept (WTA) separately by respondent demographics. See footnotes to Table 1. All estimates are expressed as percent of the wage for a job without the respective amenity. An observation is a respondent x amenity experiment pair. The sample is restricted to currently employed workers and excludes always-takers (i.e., respondents who chose always the job with amenity across all five amenity experiments) and never-takers (i.e., respondents who never chose the job with amenity across all five amenity experiments). The number of observations across demographic samples is the same by coincidence. The sample includes 1450 experiments for 290 women (210 Black, 80 not Black) and 530 experiments for 106 men (80 Black, 26 not Black). Standard errors are obtained by cluster bootstrap at the respondent level (1,000 replications). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.7: Within-respondent estimates of relative monetary values, relative to omitted amenity

	(1) UI	(2) Leave	(3) Notice	(4) Learning	(5) Commute
Unemployment insurance	-	0.0992*** (0.0276)	0.158*** (0.0243)	0.0542* (0.0279)	0.221*** (0.0297)
Parental leave	-0.110*** (0.0344)	-	0.0655*** (0.0245)	-0.0500 (0.0314)	0.135*** (0.0313)
Termination notice	-0.188*** (0.0345)	-0.0701** (0.0280)	-	-0.124*** (0.0298)	0.0746*** (0.0278)
Learning opportunity	-0.0573* (0.0312)	0.0476* (0.0284)	0.110*** (0.0236)	-	0.176*** (0.0267)
Shorter commute	-0.284*** (0.0483)	-0.156*** (0.0409)	-0.0806** (0.0322)	-0.214*** (0.0390)	-
Observations	2170	2170	2170	2170	2170

Note: This table shows conditional logit estimates including a respondent fixed effect (and omitting one amenity, one column at a time). Each RMV estimate should be interpreted as relative to the RMV for its respective omitted category. Columns (2)-(6) report relative monetary values omitting one amenity at a time. Standard errors are obtained by cluster bootstrap at the respondent level (1,000 replications). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.8: RMV, WTP, and WTA *including always-takers and never-takers*

	(1)	(2)	(3)
	RMV	WTP	WTA
Unemployment insurance	0.409*** (0.0222)	0.349*** (0.0284)	0.505*** (0.0453)
Parental leave	0.339*** (0.0215)	0.318*** (0.0274)	0.365*** (0.0286)
Termination notice	0.298*** (0.0185)	0.201*** (0.0314)	0.385*** (0.0352)
Learning opportunity	0.373*** (0.0202)	0.347*** (0.0311)	0.371*** (0.0348)
Shorter commute	0.225*** (0.0241)	0.0846* (0.0457)	0.297*** (0.0335)
Observations	3245	2930	2930

Note: This table shows relative monetary values (RMV), as percent of the wage for the job without the amenity, for five job amenities. Column (1) shows estimates for the full sample. Column (2) shows estimates for formal workers. Column (3) shows estimates for informal workers. Column (4) shows estimates for self-employed workers. RMV represents the wage gain that would make a respondent indifferent between a job without the amenity and a job with the amenity. An observation is a respondent x amenity experiment pair. RMV estimates are derived from preference parameters estimated in a logit regression where the outcome variable is a dummy for whether the respondent chose the job with the amenity, and the explanatory variables are the log wage ratio between Job B (with amenity) and Job A (baseline) and five amenity dummies. The sample includes all respondents, including always-takers (i.e., respondents who always chose the job with amenity across all five amenity experiments) and never-takers (i.e., respondents who never chose the job with amenity across all five amenity experiments). Standard errors are obtained by cluster bootstrap at the respondent level (1,000 replications) * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.9: Preference parameters used in RMV estimation

	(1)	(2)	(3)	(4)	(5)
	All	Formal	Informal	Self-employed	Unemployed
Chooses job with amenity					
$\ln(w^1/w^0)$	4.058*** (0.366)	4.511*** (0.612)	3.452*** (0.597)	4.054*** (0.807)	4.494*** (1.549)
Termination notice	0.821*** (0.117)	1.348*** (0.192)	0.330 (0.203)	0.0930 (0.285)	1.743*** (0.452)
Parental leave	1.132*** (0.119)	1.265*** (0.189)	1.151*** (0.213)	0.725*** (0.273)	1.253*** (0.427)
Learning opportunity	1.363*** (0.119)	1.422*** (0.196)	1.032*** (0.204)	1.275*** (0.285)	3.196*** (0.665)
Unemployment insurance	1.659*** (0.134)	2.368*** (0.233)	1.471*** (0.223)	0.376 (0.268)	2.839*** (0.593)
Shorter commute	0.511*** (0.104)	0.583*** (0.172)	0.657*** (0.187)	0.750*** (0.258)	-0.853** (0.387)
Observations	2170	885	705	390	190

Note: This table shows preference parameters estimated from logit regressions. Column (1) shows estimates for the full sample. Columns (2)-(5) show estimates for formal, informal, self-employed, and unemployed workers, respectively. The outcome variable is a dummy for whether the respondent chose the job with the amenity in each experiment. The explanatory variables, listed as rows, are the log wage ratio and five amenity dummies (one for each of the five experiments: unemployment insurance, parental leave, termination notice, learning opportunity, and shorter commute). Job A mirrors the respondent's current or most recent job, while Job B differs only in the specified amenity and wage. An observation is a respondent x amenity experiment pair. The coefficient on the log wage ratio (β) captures the marginal indirect utility of wages, while the amenity coefficients (δ) capture the indirect utility from each amenity. The sample excludes always-takers and never-takers. Standard errors in parentheses are clustered by respondent. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.10: RMV estimates: Logit vs OLS

	(1)	(2)	(3)	(4)
	Logit (switchers)	Logit (all)	OLS (switchers)	OLS (all)
Unemployment insurance	0.336*** (0.0222)	0.409*** (0.0222)	0.645*** (0.0233)	0.711*** (0.0238)
Parental leave	0.243*** (0.0226)	0.339*** (0.0215)	0.594*** (0.0257)	0.679*** (0.0251)
Termination notice	0.183*** (0.0179)	0.298*** (0.0185)	0.558*** (0.0238)	0.658*** (0.0253)
Learning opportunity	0.285*** (0.0192)	0.373*** (0.0202)	0.619*** (0.0239)	0.696*** (0.0236)
Shorter commute	0.118*** (0.0245)	0.225*** (0.0241)	0.518*** (0.0290)	0.614*** (0.0311)
Observations	2170	3245	2170	3245

Note: This table compares relative monetary value estimates, as percent of the job without amenity, for five job amenities using different estimation methods and samples. Columns (1) and (2) show logit estimates: Column (1) excludes always-takers and never-takers (switchers only), while Column (2) includes all respondents. Columns (3) and (4) show OLS estimates with the same sample restrictions. An observation is a respondent x amenity experiment pair. Always-takers are respondents who always chose the job with amenity across all five amenity experiments; never-takers are respondents who never chose the job with amenity across all five amenity experiments. RMV estimates in all columns are derived from preference parameters estimated using the respective method (logit or OLS). Standard errors are obtained by cluster bootstrap at the respondent level (1,000 replications). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

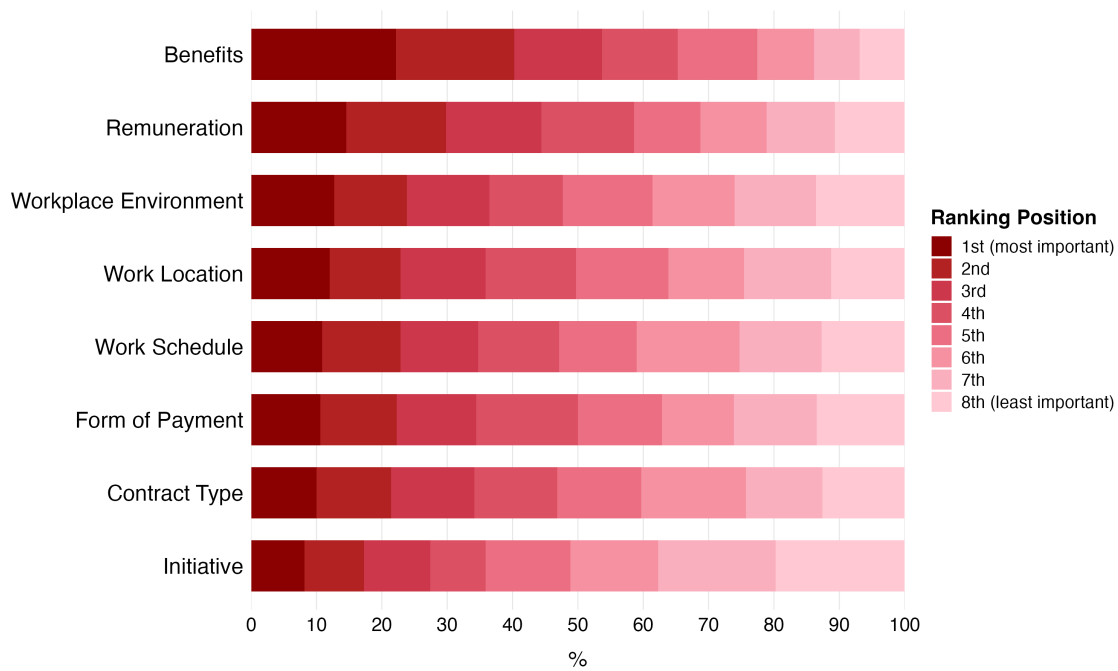
Table A.11: Preference parameters for RMV estimation: Logit vs OLS

	(1)	(2)	(3)	(4)
	Logit (switchers)	Logit (all)	OLS (switchers)	OLS (all)
$\ln(w^1/w^0)$	4.058*** (0.366)	3.931*** (0.317)	0.827*** (0.0653)	0.740*** (0.0510)
Termination notice	0.821*** (0.117)	1.393*** (0.104)	0.675*** (0.0253)	0.794*** (0.0191)
Parental leave	1.132*** (0.119)	1.627*** (0.106)	0.746*** (0.0251)	0.841*** (0.0190)
Learning opportunity	1.363*** (0.119)	1.837*** (0.109)	0.798*** (0.0250)	0.881*** (0.0191)
Unemployment insurance	1.659*** (0.134)	2.065*** (0.121)	0.857*** (0.0249)	0.917*** (0.0189)
Shorter commute	0.511*** (0.104)	1.001*** (0.0903)	0.604*** (0.0223)	0.704*** (0.0169)
Observations	2170	3245	2170	3245

Note: This table shows preference parameters estimated from logit versus OLS regressions, separately by whether the sample includes all respondents, columns (2) and (4), and if the sample excludes always-takers and never-takers of the job with the experiment-specific amenity, columns (1) and (3). The outcome variable is a dummy for whether the respondent chose the job with the amenity in each experiment. The explanatory variables, listed as rows, are the log wage ratio and five amenity dummies (one for each of the five experiments: unemployment insurance, parental leave, termination notice, learning opportunity, and shorter commute). Job A mirrors the respondent's current or most recent job, while Job B differs only in the specified amenity and wage. An observation is a respondent x amenity experiment pair. The coefficient on the log wage ratio (β) captures the marginal indirect utility of wages, while the amenity coefficients (δ) capture the indirect utility from each amenity. The sample excludes always-takers and never-takers. Standard errors in parentheses are clustered by respondent. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

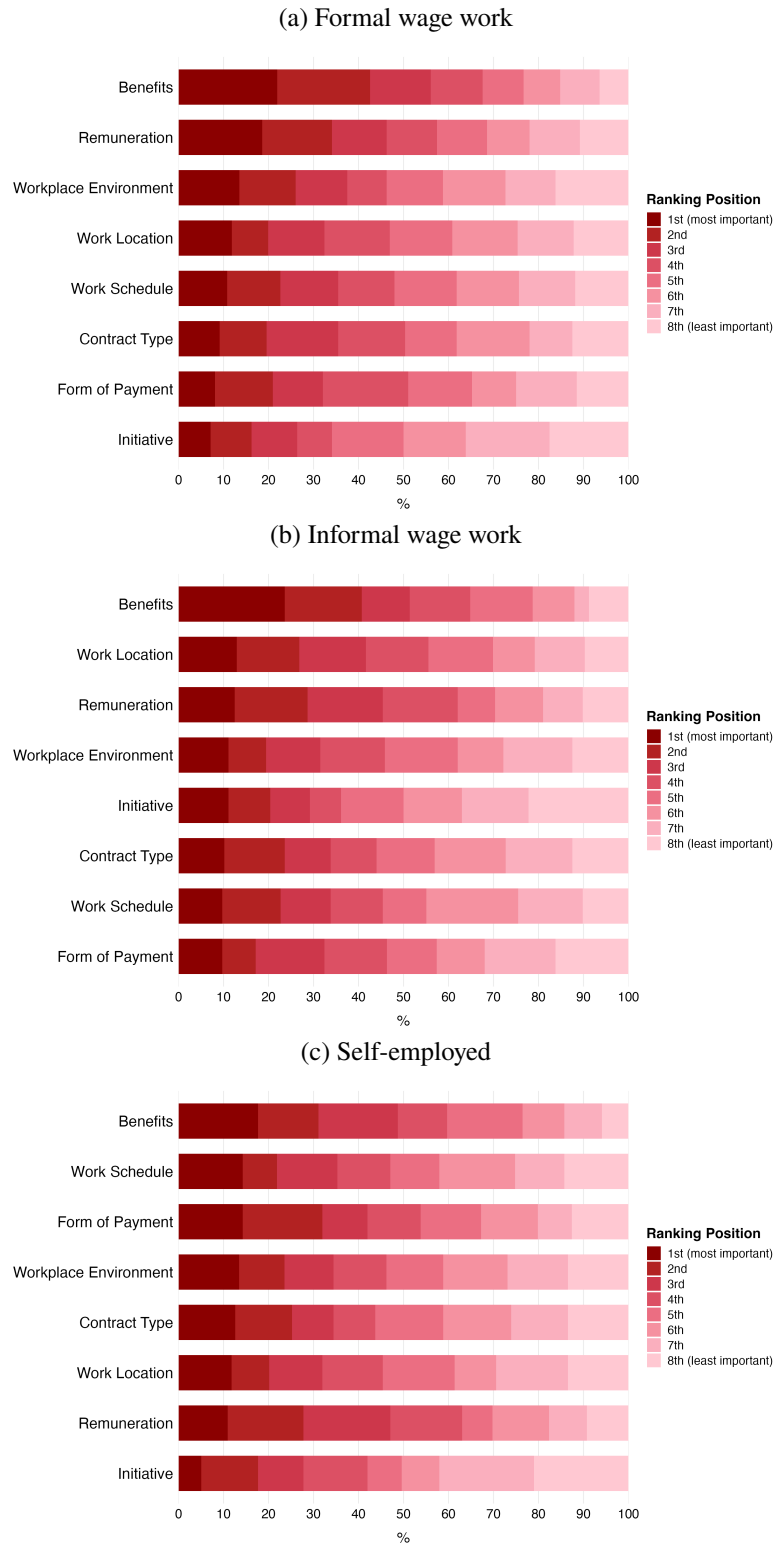
Online Appendix B: Supplementary Figures

Figure B.1: Ranking distribution of job amenities



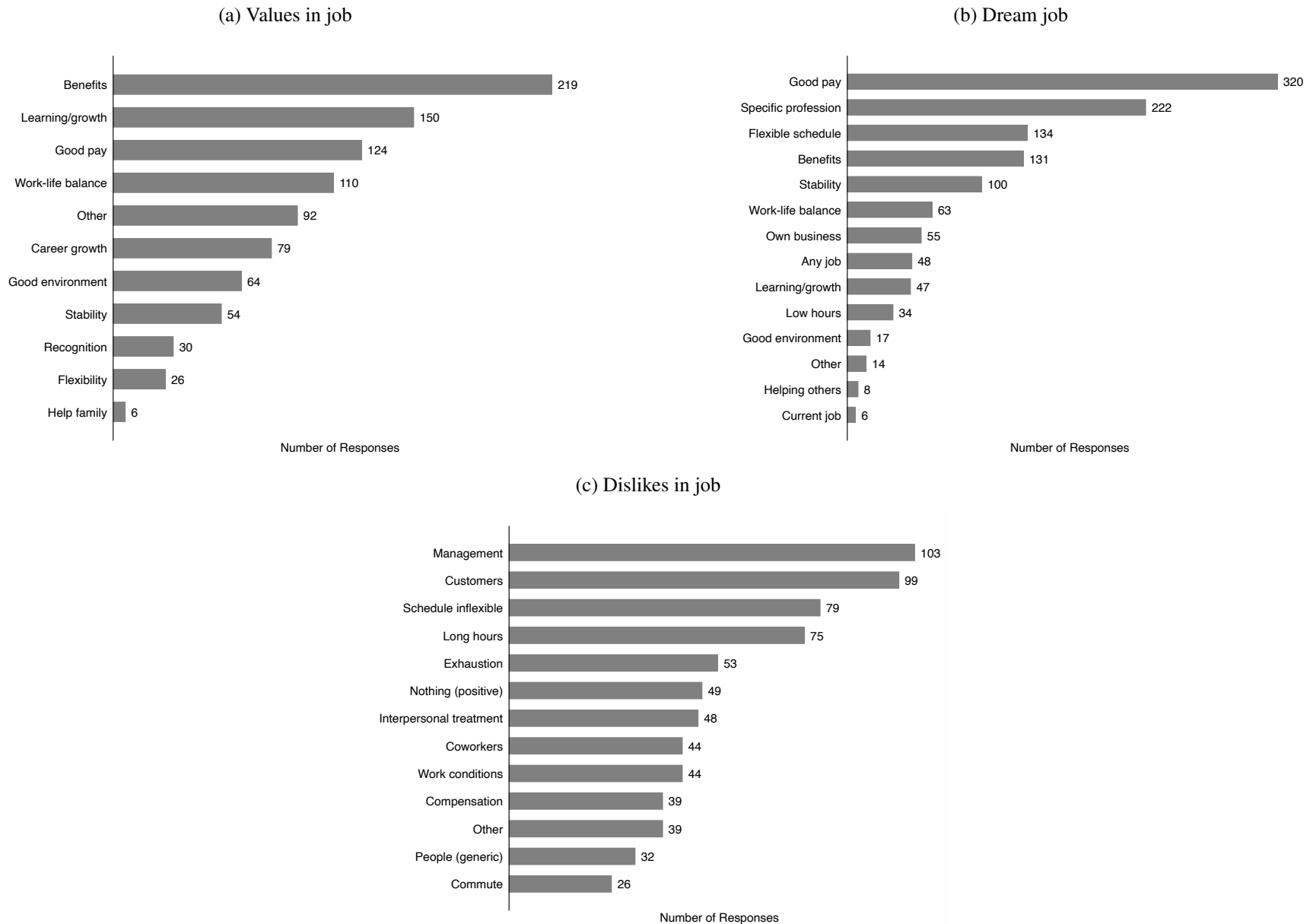
Note: This figure shows the distribution of rankings from responses to the question: “Please rank the following job amenities from most important (1) to least important (8): Benefits, Remuneration, Workplace Environment, Work Location, Work Schedule, Form of Payment, Contract Type, and Initiative.” Each stacked bar shows the percentage of respondents who assigned each ranking position to that attribute, with darker shades indicating higher importance rankings (1st = most important, 8th = least important). Based on N=700 responses.

Figure B.2: Ranking distribution of job amenities by respondent employment type



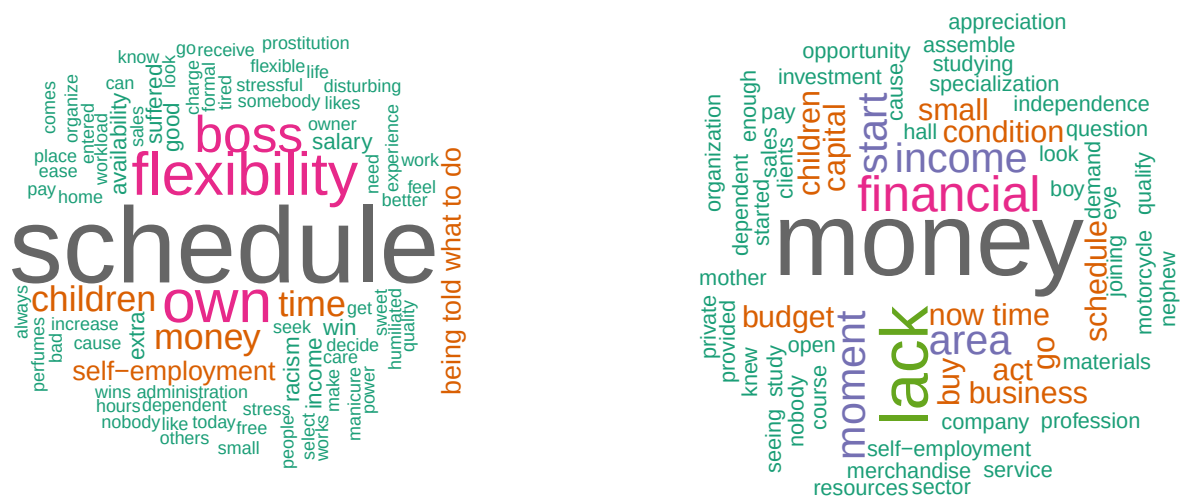
Note: This figure shows the distribution of rankings from responses to the question: “Please rank the following job amenities from most important (1) to least important (8): Benefits, Remuneration, Workplace Environment, Work Location, Work Schedule, Form of Payment, Contract Type, and Initiative,” disaggregated by respondent employment type. Each stacked bar shows the percentage of respondents who assigned each ranking position to that attribute, with darker shades indicating higher importance rankings (1st = most important, 8th = least important). Panel (a) shows formal wage workers, Panel (b) shows informal wage workers, and Panel (c) shows self-employed workers. Based on N=700 responses.

Figure B.3: Frequency distribution of open-text response categories



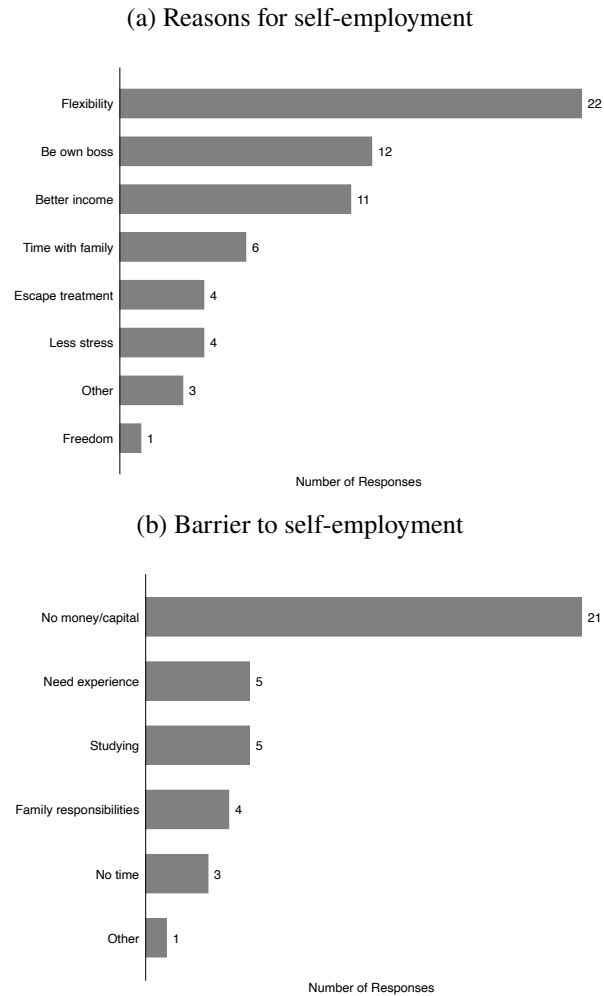
Note: This figure shows the distribution of coded categories from responses to three open-text questions. Panel (a) shows responses to “Besides salary, what else do you look for in a job?” (N=700). Panel (b) shows responses to “What would your dream job have or offer you?” (N=700). Panel (c) shows responses to “What did you like least about your most recent job?” (N=631). Categories are not mutually exclusive; respondents could mention multiple themes. Numbers on bars indicate frequency of responses mentioning each category. See Appendix C.3 for detailed category definitions and examples.

(b) Barriers to self-employment



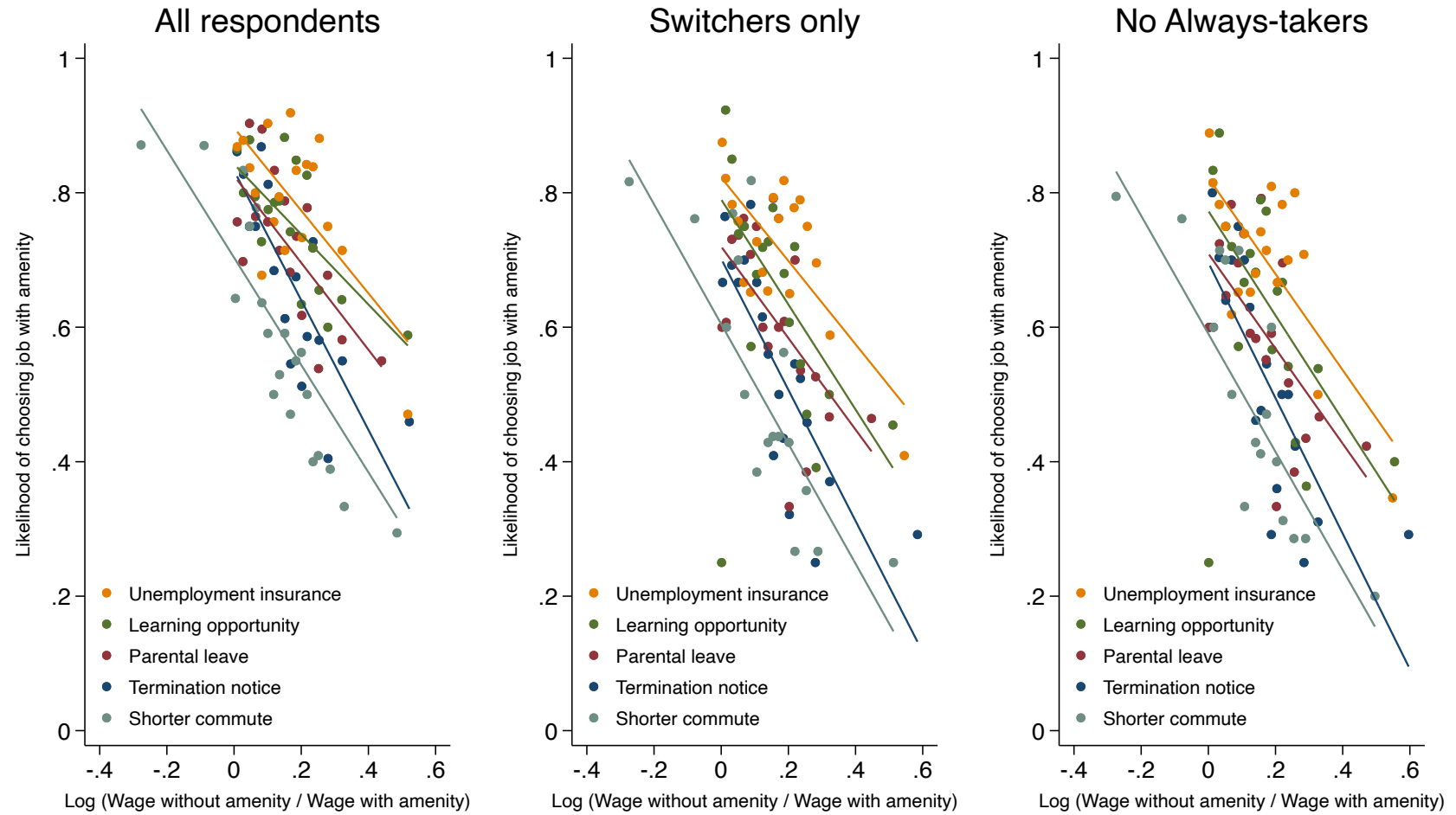
32

Figure B.5: Frequency distribution of open-text response categories



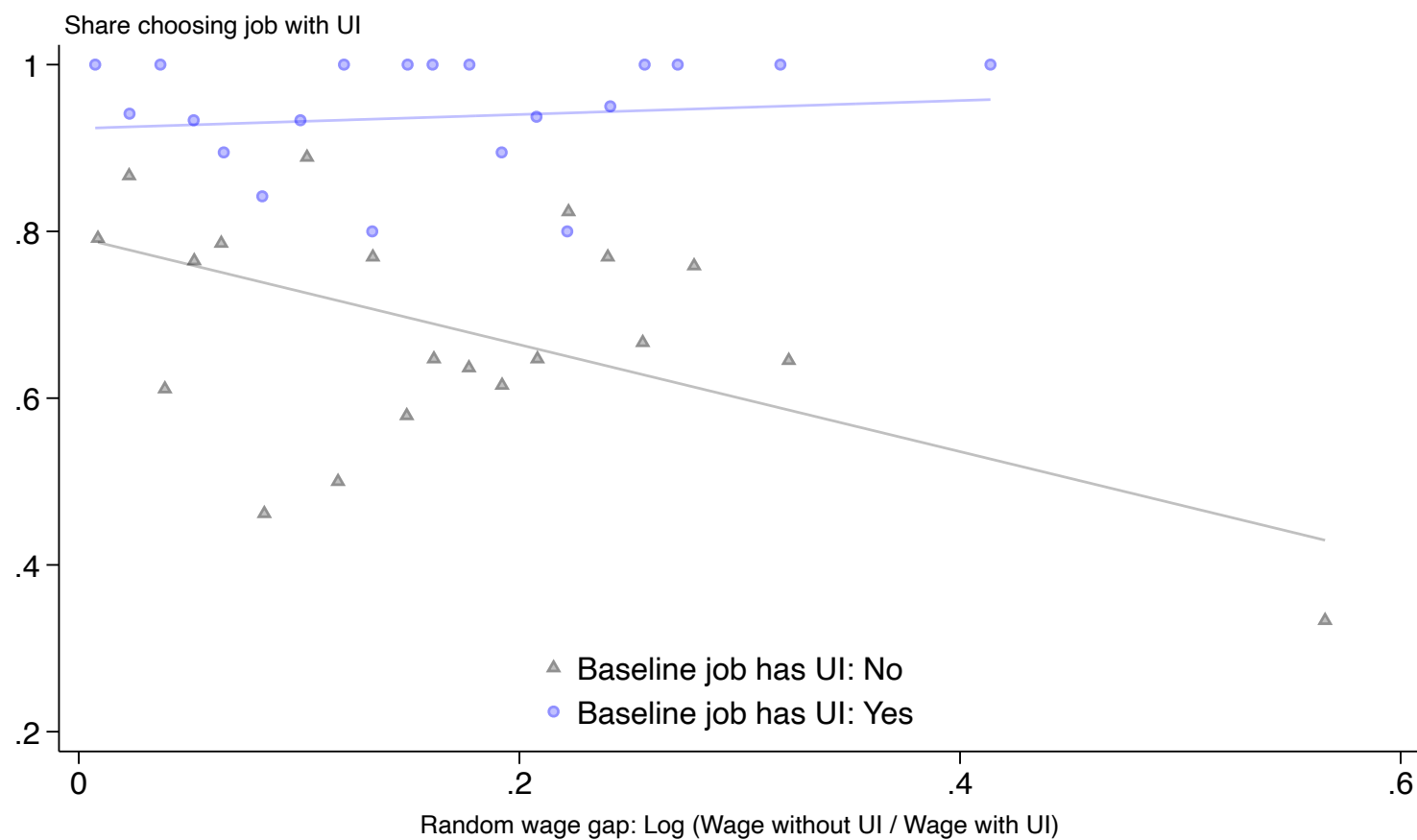
Note: This figure shows the distribution of coded categories from responses to three open-text questions. Panel (a) shows responses to “Which reasons make you want to be self-employed?” (N=45), and Panel b shows responses to “For what reasons would you rather be self-employed?” or “For what reasons are you not self-employed today?” (N=32). Categories are not mutually exclusive; respondents could mention multiple themes. Numbers on bars indicate frequency of responses mentioning each category. See Appendix C.3 for detailed category definitions and examples.

Figure B.6: Demand for job amenities by respondent type



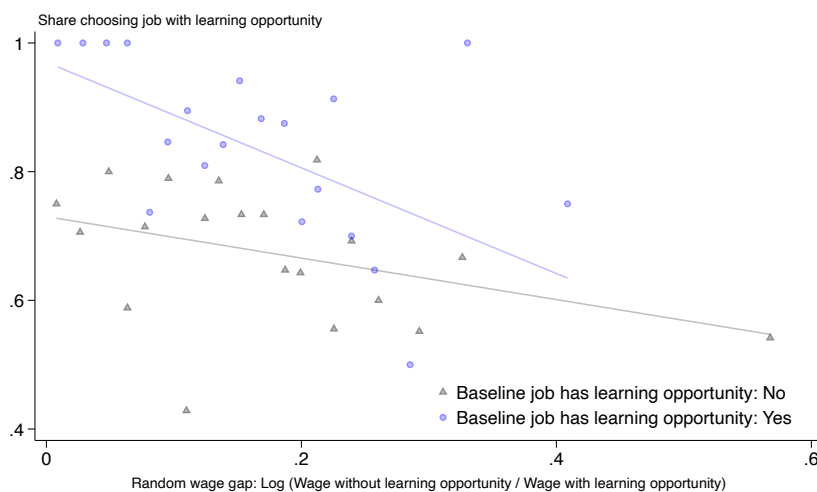
Note: This figure shows...

Figure B.7: Share of respondents choosing job with unemployment insurance

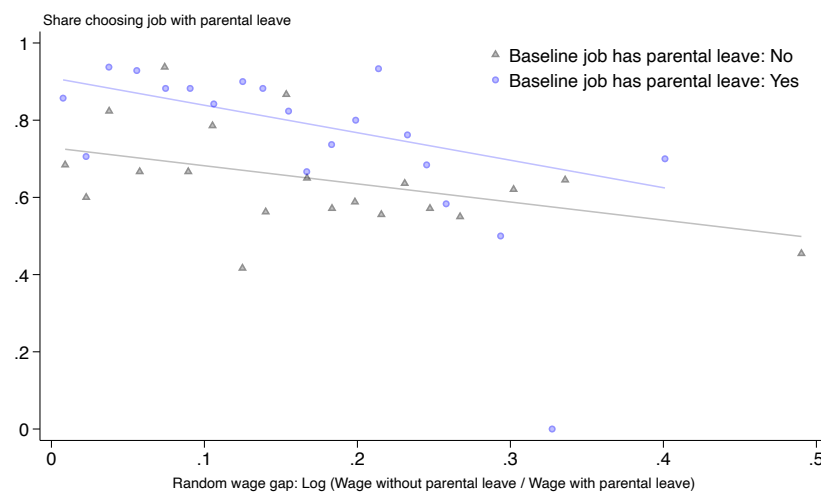


Note: This figure shows...

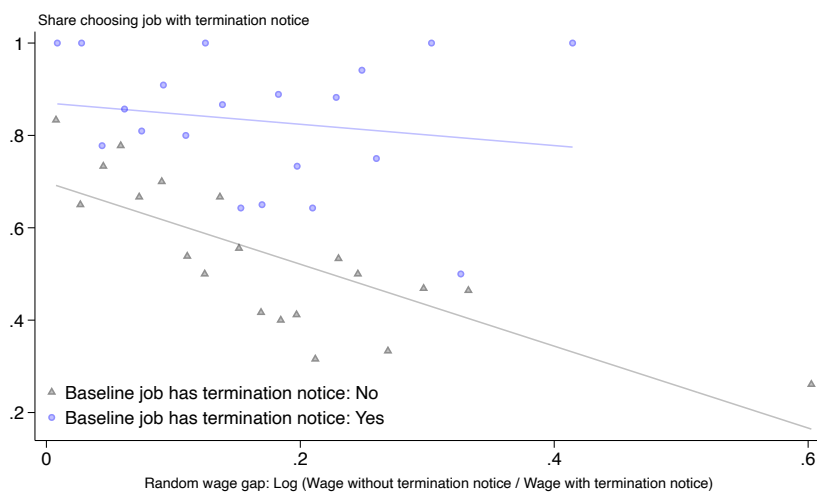
Figure B.8: Share of respondents who chose job with amenity, as function of randomly assigned wage gap, separately by amenity and by whether respondent had the amenity in their baseline job.



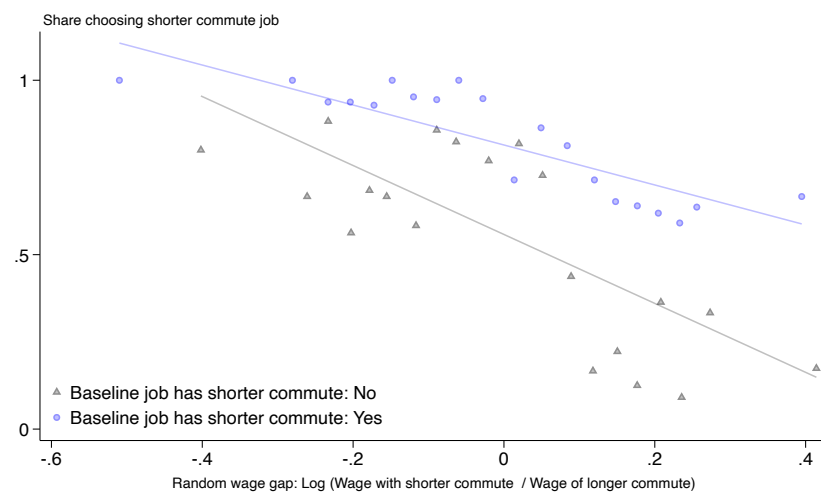
(a) Learning opportunity



(b) Parental leave



(c) Termination notice



(d) Shorter commute

Note: This figure shows...

Online Appendix C: Supplementary Materials

C.1 Survey Questions

Hello, we would like to invite you to participate in a survey about the labor market preferences of young Brazilians. The survey will ask about your demographic profile, socioeconomic context, work experience and satisfaction, as well as your preferences regarding job characteristics.

[Interviewers give consent form for participants to read and decide on their participation]

Q1. How old are you?

Q2. In the last week, did you work for pay (that is, receiving money) for at least 1 hour or take any steps to find work?

- ☐ I worked and also looked for work
- ☐ I worked, but did not look for work
- ☐ I did not work, but looked for work
- ☐ I neither worked nor looked for work

Q3. Have you ever worked before? [Only for those who indicated in question 2 that they did not work but looked for employment]

- ☐ Yes, between 1 and 6 months ago
- ☐ Yes, between 6 months and 1 year ago
- ☐ Yes, between 1 and 2 years ago
- ☐ Yes, more than 2 years ago
- ☐ No, I have never worked

Q4. In your main job, that is, the one to which you dedicate or dedicated most of your time, are/were you: [Only for those currently working or who have worked before]

- ☐ Private sector employee
- ☐ Domestic worker
- ☐ Self-employed or employer/owner
- ☐ Public sector employee
- ☐ Youth apprentice
- ☐ Intern

Q5. Does or did your job have a signed work card (*carteira assinada*)? [Only for those who indicated in question 5 that they work as: Private sector employee; Domestic worker; Public sector employee, Youth Apprentice or Intern]

☐ Yes

☐ No

Q6. Do you have MEI or CNPJ registration? [Only for those who indicated in question 5 that they are self-employed; MEI/CNPJ refer to formal registration]

☐ Yes

☐ No

Q7. What type of work are you looking for? Select all that apply. [Only for those who are looking for work]

☐ Formal employment with signed contract (CLT)

☐ Internship

☐ Youth apprentice

☐ Gigs/Self-employment

☐ Public sector

Q8. Have you ever had a job with a signed work card in the past? [Only for those who indicated in question 5 that they are self-employed and in question 6 that they do not have/did not have a signed work card]

☐ Yes

☐ No

Q9. What is the main reason for not having MEI or CNPJ registration currently? [Only for those who are self-employed and are not formally registered]

☐ Customers do not require it

☐ Costs and taxes are too high

☐ I do not know how to register for MEI or CNPJ

☐ My business does not qualify for MEI or CNPJ

☐ It takes too long to register for MEI or CNPJ

☐ I do not have all the required documents

☐ I do not want to be inspected

- ☐ I believe I may lose some social benefit if I register for MEI or CNPJ
- ☐ I never considered registering for MEI or CNPJ
- ☐ I do not know what MEI or CNPJ is
- ☐ Other. Please specify: _____

Now we will ask some questions about your preferences and opinions about jobs.

Q10. What would your dream job have or offer you? Mention some characteristics.

Q11. You indicated that you would like to be self-employed. For what reasons would you rather be self-employed? [Only for those who said they are looking for self-employment]

Q12. You indicated that you would like to be self-employed. For what reasons are you not self-employed today? [Only for those who said they are not self-employed but are looking for self-employment]

Q13. You indicated that you are working but recently looked for employment. What are the reasons that made you seek other work? [Only for those who indicated in question 3 that they worked but also looked for work]

Q14. Besides salary, what else do you look for in a job?

Q15. What do you like least about your current [last] job? [Only for those who are currently working or have worked before]

Q17. Dealing with people is part of the work environment. In your [last] work experience, is it more difficult to deal with: [Only for those who are currently working or have worked before]

- ☐ Supervisors, bosses or managers
- ☐ Coworkers
- ☐ Customers

Q19. Below, we present some characteristics related to jobs. Read carefully and rank them in order of priority for you when choosing a job.

1, which will be at the top, is the “MOST important category”

8, which will be last, is the “LEAST important category”

Remember to rank all characteristics!

- ☐ **Contract type:** Legal type of contract, such as whether it is CLT, or MEI, or no contract (autonomous).

- ☐ **Work schedule:** Flexibility, number of hours worked, shift.
- ☐ **Payment method:** Here we are not thinking about salary amount, but for example whether it is monthly and fixed, or proportional to the person's performance in the period.
- ☐ **Work location:** For example, possibility of working from home, commute time from home to work.
- ☐ **Benefits:** Whether there is meal allowance, 13th salary, termination notice, vacation, etc.
- ☐ **Work environment:** Friendships, communication, diversity, amenities.
- ☐ **Initiative:** Being in an environment that encourages initiative and proactivity.
- ☐ **Remuneration:** The amount you receive for work.

Q20. Imagine that you are offered a job with the following characteristics:

- unemployment insurance
- termination notice
- maternity leave of 4 months and paternity leave of 5 consecutive days
- commute from your home to the workplace between 1 and 2 hours
- learning opportunity

What is the minimum salary you would need to receive to accept working in this position? [Only for those who have never worked]

Q21. In your opinion, what labor rights provided by law apply to those who work with a signed work card?

[Q22-Q29 only for those who are currently working]

Q22. How many jobs or gigs did you do in the last week?

- ☐ 1
- ☐ 2
- ☐ 3 or more

Q23. From now on we will ask questions about your main job, that is, the one on which you spend the most hours if you have more than one job.

In which category of activities below does your main job fit?

- ☐ Administrative (includes administrative assistant, assistant, helper, etc.)

- ☐ Custodial services
- ☐ Attendant/salesperson
- ☐ Recyclable materials collector
- ☐ Street vendor
- ☐ Construction (mason, painter, electrician, plumber, etc.)
- ☐ Cook or pastry chef (includes kitchen assistant, chef, helper, etc.)
- ☐ Education (includes teacher, educator, daycare assistant, monitor, etc.)
- ☐ Domestic employee OR caregiver (day worker, caregiver for elderly or children, etc.)
- ☐ Car watcher
- ☐ Shop owner
- ☐ App or taxi driver
- ☐ Motorcycle taxi/Motorcycle courier/Delivery person
- ☐ Beauty sector (manicurist, hairdresser, barber, makeup artist, cosmetics sales)
- ☐ Other: _____

Q24. When did you start your current main job? You can give an approximate answer if you do not remember exactly.

Month: _____ Year: _____

Q25. What is the monthly salary you normally receive from your main job? Consider the amount that goes into your account or that you receive in cash.

Q26. Do you currently receive other sources of income?

- ☐ Bolsa Família
- ☐ BPC (Continuous Cash Benefit) [noncontributory pension]
- ☐ INSS pension (death, disability, or special pension) [contributory pension]
- ☐ Military pension
- ☐ Alimony
- ☐ Rental income
- ☐ Academic scholarships (e.g., PIBIC, CNPq)

- ☐ Other source of income
- ☐ No

Q27. How many hours did you work last week?

- ☐ Up to 20 hours
- ☐ More than 20h up to 30h
- ☐ More than 30h up to 36h
- ☐ More than 36h up to 44h
- ☐ More than 44h up to 50h
- ☐ More than 50h

Q28. How long does it normally take you to commute between your home and workplace?

- ☐ I do not commute (work from home)
- ☐ Up to 30 minutes
- ☐ 30 minutes to 1 hour
- ☐ 1 hour to 2 hours
- ☐ More than 2 hours

Q29. What benefits does your current job give you? [Yes/No for each]

- Unemployment insurance
- Meal or food allowance
- Paid vacation
- Termination notice
- Maternity or paternity leave
- Health plan
- Psychological support
- Learning opportunity
- Managing people

[Q30-Q39 only for those who have worked before but are not currently working]

Q30. From now on we will ask questions about the last job or source of income you had.

Why did you leave your last job?

- ☐ I chose to leave
- ☐ I was fired
- ☐ My contract was temporary and came to an end

Q31. Why did you choose to leave your job? [Only for those who indicated in question 31 that they chose to leave]

Q32. In the past, what was the maximum number of jobs or gigs you had at the same time?

- ☐ 1
- ☐ 2
- ☐ 3 or more

Q33. In your last job, in which category of activities below did your main job or source of income fit?

- ☐ Administrative (includes administrative assistant, assistant, helper, etc.)
- ☐ Custodial services
- ☐ Attendant/salesperson
- ☐ Recyclable materials collector
- ☐ Street vendor
- ☐ Construction (mason, painter, electrician, plumber, etc.)
- ☐ Cook or pastry chef (includes kitchen assistant, chef, helper, etc.)
- ☐ Education (includes teacher, educator, daycare assistant, monitor, etc.)
- ☐ Domestic employee OR caregiver (day worker, caregiver for elderly or children, etc.)
- ☐ Car watcher
- ☐ Shop owner
- ☐ App or taxi driver
- ☐ Motorcycle taxi/Motorcycle courier/Delivery person
- ☐ Beauty sector (manicurist, hairdresser, barber, makeup artist, cosmetics sales)
- ☐ Other: _____

Q34. What was the monthly salary you normally received from your main job? Consider the amount that went into your account or that you received in cash.

Q35. Besides the salary from your last job, did you receive other sources of income?

- ☐ Bolsa Família
- ☐ BPC (Continuous Cash Benefit) [noncontributory pension]
- ☐ INSS pension (death, disability, or special pension) [contributory pension]
- ☐ Military pension
- ☐ Alimony
- ☐ Rental income from properties
- ☐ Academic scholarships (e.g., PIBIC, CNPq)
- ☐ Other source of income
- ☐ No

Q36. In your last job, approximately how long did your employment relationship last?

- ☐ Less than 6 months
- ☐ Between 6 months and 1 year
- ☐ More than 1 year

Q37. In your last job, how many hours did you usually work per week?

- ☐ Up to 20 hours
- ☐ More than 20h up to 30h
- ☐ More than 30h up to 36h
- ☐ More than 36h up to 44h
- ☐ More than 44h up to 50h
- ☐ More than 50h

Q38. In your last job, how long did it normally take you to commute between your home and workplace?

- ☐ I did not commute (worked from home)
- ☐ Up to 30 minutes

- ☐ 30 minutes to 1 hour
- ☐ 1 hour to 2 hours
- ☐ More than 2 hours

Q39. Did your last job give you the right or opportunity to: [Yes/No for each]

- Unemployment insurance
- Meal or food allowance
- Paid vacation
- Termination notice
- Maternity or paternity leave
- Health plan
- Psychological support
- Learning opportunity
- Managing people

Q40. Have you lived in Maré since childhood?

- ☐ Yes
- ☐ No

Q41. Next, you will see two fictitious job openings for the same job with identical tasks and responsibilities. The positions are identical, except for the characteristics in bold in the table. Carefully analyze the characteristics of each position and indicate which of the two you would prefer. [Interviewers pass tablet to respondents. See Appendix C.2 for an example of how we present the discrete choices to respondents.]

Q42. Have you ever considered moving out of Maré?

- ☐ Yes. Where? And why? _____
- ☐ No. Why? _____

Q43. What gender do you identify with?

- ☐ Male
- ☐ Female
- ☐ Non-binary

☐ Prefer not to answer

☐ Other: _____

Q44. What is your race or color?

☐ not Black

☐ Black

☐ Pardo [mixed race]

☐ Asian

☐ Indigenous

☐ Prefer not to answer

Q45. What is your highest level of education?

☐ No schooling or incomplete elementary school

☐ Elementary school

☐ High school

☐ Technical high school

☐ Adult Education (EJA) - Elementary level

☐ Adult Education (EJA) - High school level

☐ College degree

☐ Master's or MBA

☐ Doctorate

Q46. Do you have children?

☐ Yes

☐ No

Q47. [Contact information for payment]

C.2 Discrete Choice Experiment

Figure 1: Discrete choice experiment screens: Formal sector benefits

(a) Unemployment insurance

The jobs below are identical, except for salary and unemployment insurance.

Carefully analyze the information below and indicate whether you would prefer job A or job B.

	Job A	Job B
Total monthly salary	R\$ 2200.00	R\$ 1145.31
Unemployment insurance	No	Yes
Commute time	I don't commute (work from home)	I don't commute (work from home)
Termination notice	No	No
Maternity leave (4 months)/ Paternity leave (5 consecutive days)	No	No
Learning opportunity	No	No

Remembering that unemployment insurance is a benefit that provides financial assistance for 3 to 5 months for workers with a signed work card who were dismissed without just cause.

- ☐ I prefer job A
☐ I prefer job B

(b) Termination notice

The jobs below are identical, except for salary and termination notice.

Carefully analyze the information below and indicate whether you would prefer job A or job B.

	Job A	Job B
Total monthly salary	R\$ 2200.00	R\$ 2174.30
Unemployment insurance	No	No
Commute time	I don't commute (work from home)	I don't commute (work from home)
Termination notice	No	Yes
Maternity leave (4 months)/ Paternity leave (5 consecutive days)	No	No
Learning opportunity	No	No

Remembering that termination notice is when the employer communicates in advance to the employee about the intention to terminate the employment relationship.

- ☐ I prefer job A
☐ I prefer job B

(c) Parental leave

The jobs below are identical, except for the salary and maternity/paternity leave.

Carefully analyze the information below and indicate whether you would prefer job A or job B.

	Job A	Job B
Total monthly salary	R\$ 2200.00	R\$ 2162.73
Unemployment insurance	No	No
Commute time	I don't commute (work from home)	I don't commute (work from home)
Termination notice	No	No
Maternity leave (4 months)/ Paternity leave (5 consecutive days)	No	Yes
Learning opportunity	No	No

Remembering that maternity leave or paternity leave is the paid time off period granted to workers after the birth or adoption of a child. Maternity leave is 4 months and paternity leave is 5 consecutive days.

- ☐ I prefer job A
☐ I prefer job B

Note: This figure shows example screens from the discrete choice experiment for three formal sector benefits. Each respondent completed five choice tasks total, one for each job attribute (unemployment insurance, termination notice, maternity/paternity leave, learning opportunity, and commute time), presented in random order. In each task, respondents chose between two hypothetical jobs with identical tasks and responsibilities. Job A mirrored the respondent's current or most recent job across all attributes. Job B differed from Job A only in the specified attribute (highlighted in bold) and in salary. The salary difference between Job A and Job B was randomly drawn from a uniform distribution ranging from -30% to $+30\%$ of Job A's salary. Starting November 19, 2025, we expanded this range to -60% to $+60\%$ to better capture higher valuations. Interviewers handed tablets directly to respondents, who selected their preferred job privately to minimize social desirability bias.

Figure 2: Discrete choice experiment screens: General job attributes

(a) Learning opportunity

The jobs below are identical, except for salary and learning opportunity.

Carefully analyze the information below and indicate whether you would prefer job A or job B.

	Job A	Job B
Total monthly salary	R\$ 2200.00	R\$ 1679.87
Unemployment insurance	No	No
Commute time	I don't commute (work from home)	I don't commute (work from home)
Termination notice	No	No
Maternity leave (4 months)/ Paternity leave (5 consecutive days)	No	No
Learning opportunity	No	Yes

Remembering that learning opportunity is when the employer offers activities or support for the employee to acquire new knowledge, skills and competencies.

☐ I prefer job A

☐ I prefer job B

(b) Commute time

The jobs below are identical, except for salary and commute time from your home to the workplace.

Carefully analyze the information below and indicate whether you would prefer job A or job B.

	Job A	Job B
Total monthly salary	R\$ 2200.00	R\$ 2082.16
Unemployment insurance	No	No
Commute time	I don't commute (work from home)	Up to 30 minutes
Termination notice	No	No
Maternity leave (4 months)/ Paternity leave (5 consecutive days)	No	No
Learning opportunity	No	No

☐ I prefer job A

☐ I prefer job B

Note: This figure shows example screens from the discrete choice experiment for two general job attributes not specific to formal employment. Each respondent completed five choice tasks total, one for each job attribute (unemployment insurance, termination notice, maternity/paternity leave, learning opportunity, and commute time), presented in random order. In each task, respondents chose between two hypothetical jobs with identical tasks and responsibilities. Job A mirrored the respondent's current or most recent job across all attributes. Job B differed from Job A only in the specified attribute (highlighted in bold) and in salary. The salary difference between Job A and Job B was randomly drawn from a uniform distribution ranging from -30% to $+30\%$ of Job A's salary. Starting November 19, 2025, we expanded this range to -60% to $+60\%$ to better capture higher valuations. Interviewers handed tablets directly to respondents, who selected their preferred job privately to minimize social desirability bias.

C.3 Open-Ended Questions

We employed AI-assisted qualitative coding to analyze the open-text questions. We used Claude AI (Anthropic, Sonnet 4.5) with extensive human oversight and iterative refinement. This approach combined the efficiency of automated text analysis with the nuance and domain expertise of human judgment. Tables C.1 through C.5 present detailed category definitions, example responses, and frequencies for each variable.

After the initial coding round, we reviewed frequency distributions to identify over- or under-represented categories and examined random samples of coded responses per category to check accuracy. When we identified miscoded responses, we provided explicit corrections to the AI, providing explicit corrections for any miscategorizations (e.g., "Response X should be coded as Y, not Z" or "Split category A into

subcategories A1 and A2 based on this distinction”). The final coding achieved coverage rates of 86.9%-98.0% across variables, with only 2.0%-13.1% of responses categorized as “other.” Responses could be assigned to multiple categories, as respondents often mentioned several aspects in a single open-text response.

Table C.1: Worst Aspects of Work (N=631)

Category	Description	Example Response	N (%)
Management	Boss/supervisor issues AND organizational problems (multiple functions, lack of support, excessive demands)	"Bad management by coordinators with accumulation of job functions" "Complete lack of dialogue and communication with supervisors"	103 (16.3)
Customers	Difficult customers, clients, patients, or dealing with the public	"Annoying and difficult customers every day" "Difficult patients who don't follow medical advice"	99 (15.7)
Schedule inflexible	Working weekends, holidays, nights, early mornings, or irregular schedules	"Having to work on weekends and holidays" "The 6x1 schedule with only one day off per week"	79 (12.5)
Long hours	Excessive work hours, overtime, or very long shifts	"The workload was extremely high with too many hours" "Constant overtime and extra hours beyond my shift"	75 (11.9)
Exhaustion	Physical and mental demands, stress, heavy workload	"Physical and mental exhaustion from the job demands" "Heavy lifting and physical effort all day long"	53 (8.4)
Nothing (positive)	Positive responses indicating satisfaction with job	"I like everything about my job" "I love my job and wouldn't change anything"	49 (7.8)
Interpersonal treatment	Mistreatment, discrimination, lack of respect, toxic environment	"Lack of respect from supervisors and management" "I experienced racism and discrimination at work"	48 (7.6)
Coworkers	Issues with colleagues, team dynamics, workplace conflicts	"Problems with coworkers who didn't cooperate" "Cliques and exclusive groups that excluded others"	44 (7.0)
Work conditions	Physical environment (noise, standing, weather), or specific disliked tasks	"Standing for long periods of time all day" "Having to clean bathrooms and dirty facilities"	44 (7.0)
Compensation	Low salary level, late payments, or lack of benefits	"The salary was far too low for the work required" "Payment was always delayed by weeks"	39 (6.2)
Other	Job instability, lack of formal contract, and miscellaneous concerns	"Not having a formal employment contract signed" "The instability and lack of security in the position"	39 (6.2)
People (generic)	Generic complaints about "people" without specifying who	"The people I had to deal with" "Having to deal with people"	32 (5.1)
Commute	Distance to work, transportation issues, or traffic	"Was too far from my home" "Spending many hours on public transportation"	26 (4.1)

Notes: This table shows the categories and example responses for the open-text question "What did you like least about your most recent job?" Categories are not mutually exclusive; respondents could mention multiple aspects (19.5% mentioned two or more). Percentages are calculated among 631 non-null responses. Examples are English translations of Portuguese responses.

Table C.2: Dream Job Attributes (N=700)

Category	Description	Example Response	N (%)
Good pay	Good salary level, remuneration, ability to support family (excluding "fixed salary" which indicates stability)	"A good salary that allows me to live comfortably" "To earn well and support my family's needs"	320 (45.7)
Specific profession	Mentions a specific job, occupation, or professional field	"I would like to work as a doctor in healthcare" "To be a chef in the kitchen, which I love"	222 (31.7)
Flexible schedule	Flexible hours, Monday-Friday schedule, no weekends, ability to choose shifts	"Flexible schedule so I can manage my time" "Working Monday to Friday without weekends"	134 (19.1)
Benefits	Formal employment benefits (food voucher, health insurance, signed contract)	"Food voucher and health insurance coverage" "Benefits like transportation voucher and dental plan"	131 (18.7)
Stability	Job security, employment stability, "fixed salary"	"Job stability and security for the future" "A fixed salary I can count on every month"	100 (14.3)
Work-life balance	Time with family, quality of life, comfortable location, peace	"Quality of life and time for my personal life" "More time to spend with my children"	63 (9.0)
Own business	Working for oneself, having own business or being independent	"To work for myself and have my own business" "Being independent and working on my own"	55 (7.9)
Any job	Just wants any employment without specific requirements	"Any job would work for me right now" "I just need a job, any job is fine"	48 (6.9)
Learning/growth	Opportunities to learn and grow within the job (not studying for a profession)	"Growth opportunities within the company" "Opportunities for professional development"	47 (6.7)
Low hours	Reduced total working hours, not wanting long workdays	"Few hours of work per day, not too many" "Not having to work long hours every day"	34 (4.9)
Good environment	Positive work atmosphere, recognition, respect, good teamwork	"A good work environment with nice people" "Recognition and respect for my contributions"	17 (2.4)
Other	Miscellaneous responses including purpose, satisfaction, or unclear	"A sense of purpose in what I do" "Prosperity and visibility in my field"	14 (2.0)
Helping others	Social impact, helping people, care work	"To help people who need assistance" "Saving lives through my profession"	8 (1.1)
Current job	Already satisfied with current employment	"This is already my dream job" "I struggled to get this job and it's perfect"	6 (0.9)

Notes: This table shows the categories and example responses for the open-text question "What would your dream job offer you?" Categories are not mutually exclusive; respondents could mention multiple attributes (54.7% mentioned two or more). All 700 respondents provided answers. Examples are English translations of Portuguese responses.

Table C.3: What Respondents Value Most in a Job (N=700)

Category	Description	Example Response	N (%)
Benefits	Formal employment benefits (food voucher, health insurance, transportation)	"Health insurance coverage for me and my family" "Having a formal contract with worker protections"	219 (31.3)
Learning/growth	Opportunities for learning, gaining experience, and personal development	"Learning new skills and gaining experience" "Gaining knowledge and developing myself"	150 (21.4)
Good pay	Adequate salary, income, or financial independence	"Just the salary itself is most important" "Financial independence and good income"	124 (17.7)
Work-life balance	Quality of life, reasonable hours, time management, peace	"Quality of life and work-life balance" "Few hours so I have time for my family"	110 (15.7)
Other	Miscellaneous values that don't fit other categories	"Just having any job is what matters" "Being able to maintain myself is enough"	92 (13.1)
Career growth	Professional advancement opportunities, career trajectory, promotions	"Professional growth within the company" "Growing inside the organization over time"	79 (11.3)
Good environment	Positive work atmosphere, respect, organization, good colleagues	"A good work environment with pleasant people" "Working with a good team of colleagues"	64 (9.1)
Stability	Job security and employment stability	"Stability and security in my position" "Knowing I have stable employment"	54 (7.7)
Recognition	Recognition of work, respect, feeling valued	"Recognition for my work contributions" "Being valued for what I do at work"	30 (4.3)
Flexibility	Schedule flexibility, accessibility, easy access	"Flexibility in my work schedule" "Having some flexibility in how I work"	26 (3.7)
Help family	Ability to help and support family members	"Being able to help out at home financially" "Supporting my daughter with what she needs"	6 (0.9)

Notes: This table shows the categories and example responses for the open-text question "What do you value most in a job?" Categories are not mutually exclusive; respondents could mention multiple values. All 700 respondents provided answers. Examples are English translations of Portuguese responses.

Table C.4: Reasons for Wanting Self-Employment (N=45)

Category	Description	Example Response	N (%)
Flexibility	Schedule flexibility and ability to make own hours	"Flexibility in my schedule to manage my time" "Having flexible hours because of my children"	22 (48.9)
Own boss	Being own boss, not having a supervisor, independence	"I want to be my own boss and leader" "Not having a boss telling me what to do"	12 (26.7)
Income	Better income potential, extra money, financial opportunity	"To earn better money than as an employee" "I can increase my earnings working for myself"	11 (24.4)
Family time	More time with children or family members	"More time to spend with my children at home" "So I have time for my family needs"	6 (13.3)
Less stress	Reduced stress, fewer demands, less pressure	"It's less stressful than working for someone" "I would receive less pressure and demands"	4 (8.9)
Escape treatment	Escaping bad treatment, discrimination, or mistreatment	"I got tired of being humiliated at work" "I suffered racism in all my jobs as an employee"	4 (8.9)
Other	Miscellaneous reasons including unclear responses	"I still don't know how to organize myself" "I like selling perfumes for myself"	3 (6.7)
Freedom	Sense of freedom, autonomy, independence	"I feel freer working for myself" "Having personal freedom and autonomy"	1 (2.2)

Notes: This table shows the categories and example responses for the open-text question "Why do you want to work for yourself?" (asked conditionally to those who expressed interest in self-employment). Categories are not mutually exclusive; respondents could mention multiple reasons. Only 45 of 700 respondents answered this question. Examples are English translations of Portuguese responses.

Table C.5: Reasons Against Self-Employment (N=32)

Category	Description	Example Response	N (%)
No money	Lack of financial capital, money, or resources to start	"I don't have enough money to start a business" "No funds available to buy inventory or materials"	21 (65.6)
Studying	Currently studying or need to complete education first	"I'm currently studying and can't work yet" "Need to finish my specialization first"	5 (15.6)
Need experience	Need professional experience or qualification first	"I still don't know which area to work in" "I need to have a profession first"	5 (15.6)
Family responsibilities	Children or family care duties prevent self-employment	"My daughters are too young and need care" "I take care of my nephew full-time"	4 (12.5)
No time	Lack of time or organizational capacity	"I don't have time to organize a business" "Don't have the organization skills needed"	3 (9.4)
Other	Miscellaneous reasons including industry dependencies	"I already work as a motorcycle courier" "My work depends on private sector companies"	1 (3.1)

Notes: This table shows the categories and example responses for the open-text question "Why don't you want to work for yourself?" or "Why can't you work for yourself?" (asked conditionally to those who did not want self-employment). Categories are not mutually exclusive; respondents could mention multiple reasons. Only 32 of 700 respondents answered this question. Examples are English translations of Portuguese responses.

Online Appendix D: Theory Appendix

Proposition 1. *Willingness to pay and willingness to accept are the same for agents whose preferences are not reference-dependent.*

Proof. This equivalence has been shown before, as its core arguments go as far back as Hicks. Compensating variation and equivalent variation are the same under no income effects (or endowment effects in the case of reference dependence). Quasi-linear utility delivers this. We show the equivalence with our notation to help readers more easily interpret our estimates of relative monetary values, willingness to pay, and willingness to accept.

Let agents have indirect utility over jobs as posited in equation 1 of Section 4. We first show that this agent's willingness to pay to get an amenity equals Section 4 expression for relative monetary values. Then, we show the same for the agent's willingness to accept to forgo the same amenity.

First, consider a respondent who has job j^0 and let $WTP_{a^*} > 0$ be the wage *cut* that would make her indifferent between her current job and a job with the amenity, such that $w_{j^0} - WTP_{a^*} = w_{j^1}$. Plugging w_{j^1} into equation 4 and solving for WTP_{a^*} gives

$$\ln \left(\frac{w_{j^0} - WTP_{a^*}}{w_{j^0}} \right) = -\frac{\delta_a^*}{\beta} \implies WTP_{a^*} = w_{j^0} \left[1 - \exp \left(-\frac{\delta_a^*}{\beta} \right) \right] = RMV_{a^*} \quad (9)$$

Now, consider a respondent who has job j^1 and let $WTA_{a^*} > 0$ be the wage *increase* that would make her indifferent between her current job and a job without the amenity, such that $w_{j^1} + WTA_{a^*} = w_{j^0}$. Plugging w_{j^1} into equation 4 and solving for WTA_{a^*} gives

$$\ln \left(\frac{w_{j^0} - WTA_{a^*}}{w_{j^0}} \right) = -\frac{\delta_a^*}{\beta} \implies WTA_{a^*} = w_{j^0} \left[1 - \exp \left(-\frac{\delta_a^*}{\beta} \right) \right] = RMV_{a^*} \quad (10)$$